Stochastic Integral Reinforcement Learning for Distributed H_∞ Optimal Formation Tracking of Multi-Agent Wheeled Mobile Robots

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Abstract: In this paper a distributed stochastic H_∞ optimal tracking control problem for multi-agent wheeled mobile robot systems is proposed, under both Itô process noise in the agent dynamics and worst-case external disturbances. By reformulating the control task as a stochastic zero-sum game problem, a decentralized framework is established wherein each agent interacts locally with its neighbors and is influenced by disturbances. The solution involves a set of coupled stochastic Hamilton–Jacobi–Isaacs equations, which gain a genuine Itô-correction term over their deterministic counterpart and reduce to them exactly as the noise intensity vanishes. To derive an approximate optimal policy, a critic-only neural network scheme is developed. A historical data-based update law is employed to eliminate the need for persistence of excitation, thereby enhancing online learning efficiency. Furthermore, the proposed approach does not require knowledge of the drift dynamics of the augmented system. Simulation results are presented to demonstrate the effectiveness and real-time capability of the proposed method.

Keywords: Wheel mobile robot, Multi-agent system, Stochastic control, Reinforcement learning, Adaptive dynamic programming, Graphical games

1 Introduction

In recent years, cooperative control of multi-agent systems has attracted substantial attention from both the scientific and engineering communities, primarily due to its wide range of applications in areas such as traffic management, power grids, resource distribution, wireless sensor networks, and robotic swarms. Among these, formation control represents a fundamental challenge, involving the coordination of multiple autonomous agents to maintain predefined geometric patterns during motion. This problem has gained increasing importance owing to several inherent difficulties, including system heterogeneity, reliance on locally available information, model uncertainties, and the complexity of dynamic and unstructured environments[1,2,3].

A variety of formation tracking control strategies have been developed for multi-robot systems, including adaptive control[4,5,6], sliding mode control (SMC)[7,8,9], and model predictive control (MPC), among others. In [7], a high-order SMC approach is proposed for two-wheeled mobile robots, where a hyperbolic tangent-based sliding surface is adopted to mitigate the effects of external disturbances and alleviate chattering. Additionally, a radial basis function neural network (RBFNN) is employed to approximate the uncertain system dynamics. A robust fast fixed-time formation control method is presented in [8] for autonomous underwater vehicles (AUVs), using a leader–follower framework. The control design integrates fixed-time SMC and a sliding mode observer (SMO) to ensure tracking performance in the presence of disturbances and actuator faults. Similarly, a leader–follower SMC strategy is applied to quadrotor unmanned aerial vehicles (UAVs) in [9], wherein a distributed estimator is designed to compensate for intermittent communication between the leader and followers. Fractional-order SMC is then utilized to enable each follower to maintain the desired formation configuration.

Alongside SMC-based approaches, adaptive control schemes have also been widely adopted to address model uncertainties in formation control of multi-robot systems. In [4], an adaptive formation tracking method using quantized communication is proposed for networked nonholonomic mobile robots, where unknown slippage constraints are handled through an adaptive estimator. Discontinuous control signals are generated via quantized state feedback and relative positioning. In [5], a finite-time adaptive formation control framework with event-triggered communication is designed, where the unknown nonlinear dynamics of follower robots are approximated by fuzzy logic systems. Adaptive control laws are then developed to ensure finite-time convergence to the leader's trajectory. Furthermore, [6] presents an adaptive control architecture using the unscented Kalman filter (UKF) to estimate the target's motion. The estimated information is used in a backstepping-based leader–follower controller, resulting in a smoother control input and improved tracking accuracy.

In recent years, reinforcement learning (RL) has considered as a powerful framework for solving nonlinear optimal control problems. Within the context of multi-agent system (MAS) control, RL techniques have been actively investigated to enhance system-wide performance. For example, game-theoretic backstepping controllers have been analyzed for nonlinear MAS in [10], where each performance index function incorporates interactions between pairs of agents. The game-based designs demonstrated superior convergence rates compared to conventional optimal controllers. To address the optimal consensus problem in discrete-time MAS, a model-free Q-learning approach is proposed in [11], leveraging system trajectory data instead of requiring accurate system models. A data-driven RL framework is further introduced in [12] to achieve optimal robust synchronization among heterogeneous agents. In [13], an off-policy integral reinforcement learning method is proposed to resolve the optimal synchronization task without needing prior knowledge of agent dynamics. To manage optimal synchronization in MAS with actuator constraints, [14,15] enhance the actor–critic neural network paradigm by incorporating a vanishing viscosity mechanism. The H_∞/zero-sum-game route to disturbance-robust optimal control that this paper follows traces back to the classical dynamic-game formulation of [16], and the specific critic-only, synchronous-policy-iteration mechanism this paper's Section 4 builds on was first developed for deterministic nonlinear zero-sum games in [17, 18].

Collectively, these studies emphasize optimal coordination strategies in MAS using RL techniques. However, most existing RL algorithms rely on either policy iteration or value iteration, and, with few exceptions, are posed for *deterministic* agent dynamics – leaving the practically important case of agents subject to genuine process noise (actuator ripple, unmodeled dynamics, terrain-induced excitation), rather than only bounded worst-case disturbance inputs, comparatively underexplored. While policy iteration typically necessitates an initially admissible policy, value iteration is known for its slower convergence. Furthermore, the dependency on the persistence of excitation (PE) condition during the early learning phase remains a significant barrier to the practical deployment of RL algorithms in real-time applications. Despite these advancements, challenges persist in achieving decentralized optimal tracking control for multi-agent systems under both stochastic process noise and worst-case external disturbances, without full knowledge of system dynamics. To address these issues, this paper introduces a distributed stochastic H_∞ optimal tracking control framework for multi-agent wheeled mobile robot systems, in which each agent's dynamics are modeled as an Itô stochastic differential equation rather than an ordinary differential equation, in the spirit of the stochastic Hamilton–Jacobi–Bellman/Isaacs theory of [19] and the linear stochastic zero-sum-game solution of [20] that the recent model-free policy-iteration algorithm of [21] builds on. Unlike that linear, quadratic-cost setting – where the value function admits a closed-form (Riccati-type) solution – the nonholonomic wheeled-robot dynamics considered here are nonlinear, so no such closed form exists even after adding process noise; this is precisely why a critic-only neural-network solution to the resulting coupled stochastic Hamilton–Jacobi–Isaacs (HJI) equations remains necessary, and is the route this paper takes. By reformulating the control task as a stochastic zero-sum game and employing a critic-only neural network structure, the proposed approach effectively eliminates the need for persistence of excitation and prior knowledge of drift dynamics, while explicitly accounting for the diffusion term that a purely deterministic formulation omits; an off-policy reformulation then removes the remaining dependence on the input and diffusion channels, yielding a completely model-free algorithm. This contribution aims to improve learning efficiency and real-time applicability in complex and uncertain environments.

The primary contributions of this study are summarized as follows

- The distributed H_∞ optimal formation tracking problem for multi-agent nonholonomic wheeled mobile robots is formulated under *stochastic*, rather than deterministic, agent dynamics: each agent's strict-feedback dynamic subsystem is modeled as an Itô SDE with a Lipschitz, linear-growth diffusion term. The resulting coupled Hamilton–Jacobi–Isaacs (HJI) equations gain a genuine second-order (Itô-correction) term absent from the deterministic formulation, which we show reduces to the deterministic HJI equations exactly as the diffusion intensity vanishes.

- A reinforcement learning framework based on a critic-only neural network (NN) is introduced to approximate the solution of the coupled stochastic HJI equations. By integrating experience replay into an integral reinforcement learning process re-derived via Itô's formula and conditional expectation, the proposed method reduces dependence on the conventional PE condition while preserving system stability during learning. This approach enables the design of a novel adaptive law for updating NN parameters under process noise.

- The scheme is then made *completely model-free* through an off-policy reformulation of the stochastic IRL Bellman equation. By reading the stationarity conditions backwards, the unknown products $\nabla V_i^T g_i$ and $\nabla V_i^T k_i$ are lumped into the agent's own policies weighted by the design matrices R_{ii}, T_{ii}, while the agent-coupled cross-channel terms $\nabla V_i^T g_j$, which admit no such identity in the graphical-game setting, are identified from data alongside the critic weights. The resulting algorithm requires no knowledge of the drift, the input channels, or the diffusion, and – because the process noise enters additively rather than through the control channel – needs no diffusion-dependent parameter to be identified, unlike its linear stochastic counterpart [21]. The extension amounts to a change of regressor rather than a change of algorithm, so the stability guarantee carries over unchanged.

- The mean-square stability of the resulting closed-loop system is rigorously established via a stochastic Lyapunov argument, and the convergence of control and disturbance strategies to the Nash equilibrium is guaranteed in the mean-square sense, with an explicit noise-dependent ultimate bound that vanishes in the deterministic limit.

The rest of the paper is organized as follows. Section 2 provides the required preliminaries and formulates the stochastic formation tracking problem. Section 3 reformulates the problem as a stochastic zero-sum differential graphical game and derives the coupled stochastic HJI equations. Section 4 presents the critic-only online integral reinforcement learning algorithm and its mean-square stability analysis. Section 5 illustrates the performance of the proposed algorithm through simulation. Conclusions and directions for future research are given in Section 6.

2 Preliminaries and problem formulation

Notations: In this paper, the notations $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{R}^{m\times n}$ denote the set of real numbers, the n-dimensional Euclidean space, and the set of real-valued $m \times n$ matrices, respectively. The symbol $\mathbf{I}_n$ represents the n-dimensional identity matrix, while $\mathbf{1}_N$ and $\mathbf{0}$ denote the N-dimensional column vector of ones and the zero vector of appropriate dimension, respectively. The notation $\|\cdot\|$ stands for the standard Euclidean norm when applied to vectors, and the induced matrix norm when applied to matrices. The $\mathcal{L}_2$-norm of a signal $\beta(t) \in \mathcal{L}_2^n[0,\infty)$ is defined as $\|\beta(t)\|_{\mathcal{L}_2} = \left(\int\limits_0^\infty \left|\beta(t)\right|^2 d\tau\right)^{\frac{1}{2}} < +\infty$, $\alpha(t) \in \mathbb{R}^n$. The Kronecker product is denoted by $\otimes$. The notation $\mathrm{diag}\{x_1, x_2, \ldots, x_n\}$ refers to a diagonal matrix whose diagonal entries are $x_1, x_2, \ldots, x_n$. The symbol $\lambda_{\min}(M)$ indicates the smallest eigenvalue of matrix M.

2.1 Graph theory

For a MAS consisting of N follower agents, the underlying communication topology can be modeled as a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, \ldots, v_N\}$ denotes the set of nodes (agents), and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of directed edges, with $|\mathcal{E}| = m$ being the total number of edges. The adjacency matrix associated with $\mathcal{G}$ is defined as $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N\times N}$, where $a_{ij} > 0$ if there exists a directed edge from node v_j to node v_i, i.e., $(v_j, v_i) \in \mathcal{E}$, and $a_{ij} = 0$ otherwise. Notably, self-loops are excluded, i.e., $a_{ii} = 0$ for all i. The neighbor set of agent i is denoted by $\mathcal{N}i = \{v_j \mid (v_j, v_i) \in \mathcal{E}\}$, containing all agents that send information to agent i. The in-degree matrix is denoted as $\mathcal{D} = \mathrm{diag}(d_i)$, where $d_i = \sum_{j\in\mathcal{N}_i} a_{ij}$ represents the weighted in-degree of node v_i. The Laplacian matrix is then defined as $\mathcal{L} = \mathcal{D} - \mathcal{A}$, which satisfies the property that the sum of each row is zero. If the i-th follower agent receives information directly from the leader node, then $a_{i0} > 0$; otherwise, $a_{i0} = 0$. It is assumed that at least one follower satisfies $a_{i0} > 0$. The entries b_{ij} correspond to the diagonal matrix $\mathcal{B}$, where each diagonal element is given by $b_{ii} = a_{i0}$. A directed path is a finite sequence of edges of the form $(v_1, v_2), (v_2, v_3), \ldots, (v_{k-1}, v_k)$, where $v_i \in \mathcal{V}$ for all i. The graph $\mathcal{G}$ is said to be strongly connected if, for any pair of distinct nodes $v_i, v_j \in \mathcal{V}$ with $i \neq j$, there exists a directed path from v_i to v_j. In this study, we consider a directed graph with a fixed and strongly connected topology.

2.2 Mathematical model of a nonholonomic wheeled mobile robot

Consider a multi-robot system comprising n nonholonomic wheeled mobile robots (NWMRs), each represented by a unicycle-type model with two actuated wheels and a passive caster. Let r_i and l_i denote the wheel radius and half the wheelbase of robot i, respectively. The mass center is assumed to coincide with the geometric center, located midway between the wheels in the global frame xOy. The configuration of robot i is described by the state vector (x_i, y_i, θ_i), where θ_i denotes the heading angle. Following the standard Euler–Lagrange derivation for constrained mechanical systems [22], the dynamic model is given by

$$M_i(p_i)\ddot{p}_i + C_i(p_i, \dot{p}_i)\dot{p}_i + F_i(\dot{p}_i) + G_i(p_i) + \tau_{pi}(t) = B_i(p_i)\tau_i - A_i^T(p_i)\lambda_i, \quad i = 1, \ldots, N, \tag{1}$$

where $p_i \triangleq [x_i, y_i, \theta_i]^T \in \mathbb{R}^3$ denotes the position of the robot i in the earth frame, $M_i : \mathbb{R}^3 \to \mathbb{R}^{3\times 3}$ denotes a symmetric positive-definite inertia matrix, $C_i : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^{3\times 3}$ denotes the bounded centripetal and Coriolis matrix. It is noted that the Coriolis and centrifugal term vanishes, i.e. $C_i(p_i, \dot{p}_i) = 0$, when the mass center coincides with the geometric centroid of the robot. $F_i : \mathbb{R}^3 \to \mathbb{R}^3$ denotes surface friction forces, $\tau_{pi} : [t_0, \infty) \to \mathbb{R}^3$ denotes bounded unknown disturbances including unstructured unmodeled dynamics, $G_i : \mathbb{R}^3 \to \mathbb{R}^3$ is the gravitational vector, $B_i : \mathbb{R}^3 \to \mathbb{R}^{3\times 2}$ denotes the input transformation matrix, $\boldsymbol{\tau}_i \triangleq [\tau_{i,\text{right}}, \tau_{i,\text{left}}]^T \in \mathbb{R}^2$ is the control vector with the actuation torques applied on right and left wheels, respectively, $A_i^T : \mathbb{R}^3 \to \mathbb{R}^3$ is the matrix associated with the constraints, $\lambda_i \in \mathbb{R}$ is the Lagrange multiplier associated with the nonholonomic constraint. Moreover, the robot is assumed to move without lateral slip, thereby satisfying the nonholonomic constraint.

$$\dot{x}_i \sin\theta_i - \dot{y}_i \cos\theta_i = 0, \quad \forall i \in \mathbb{N}. \tag{2}$$

Wheeled mobile robots are renowned examples of nonholonomic mechanical systems. These robots adhere to the nonholonomic constraint 2, which dictates movement exclusively along directions normal to the axes of their driving wheels. This adherence ensures pure rolling and nonslipping motion. Consequently, the kinematic model of a mobile robot i can be expressed as:

$$\dot{p}_i = H_i(p_i)\eta_i, \tag{3}$$

where the vector $\eta_i \in \mathbb{R}^2$ is composed of linear and angular velocities defined as $\eta_i \triangleq [v_i, \omega_i]^T$, and $H_i(p_i) \in \mathbb{R}^{3\times 2}$ is the full-rank velocity transformation matrix. Furthermore, the nonholonomic constraint, which can be written as $A(p_i)\dot{p}_i = 0$, can be associated with (3) to achieve that $H^T(p_i)A^T(p_i) = 0$. By applying Lagrange's formulation, we can obtain the determination of the system matrices in Eq. (1) and (2). We have $M_i = \mathrm{diag}(m_i, m_i, I_i)$, $\lambda_i = -m_i(\dot{x}_i \cos\theta_i +$

$\dot{y}_i \sin\theta_i)\dot{\theta}_i$, where m_i denotes mass, I_i is the moment of inertia, $2l_i$ represents the robot width, and R_i is the wheel radius, and other matrices as follows:

$$B_i(p_i) = \frac{1}{l_i}\begin{bmatrix}\cos\theta_i & \sin\theta_i\\ \cos\theta_i & \sin\theta_i\\ -R_i & -R_i\end{bmatrix}, \quad A_i^T(p_i) = \begin{bmatrix}-\sin\theta_i\\ \cos\theta_i\\ 0\end{bmatrix}, \quad H_i(p_i) \triangleq \begin{bmatrix}\cos\theta_i & 0\\ \sin\theta_i & 0\\ 0 & 1\end{bmatrix}.$$

To facilitate the subsequent development, the dynamic equations can be expressed in terms of the internal velocity η_i using (3), (1) and the fact $H_i^T(p_i)A_i^T(p_i) = 0$ as follows:

$$\bar{M}_i(p_i)\dot{\eta}_i + \bar{C}_i(p_i, \dot{p}_i)\eta_i + \bar{F}_i(\dot{p}_i) = \bar{B}_i(p_i)\tau_i - \bar{\tau}_{di}, \tag{4}$$

where $\bar{M}_i(p_i) \triangleq H_i^T(p_i)M_i(p_i)H_i(p_i), \bar{C}_i(p_i, \dot{p}_i) \triangleq \mathbf{0}, \bar{F}_i(\dot{p}_i) \triangleq H_i^T(p_i)F_i(\dot{p}_i), \bar{B}_i(p_i) \triangleq H_i^T(p_i)B_i(p_i)$ and $\bar{\tau}_{di} \triangleq H_i^T(p_i)\tau_{di}$. All stochastic quantities in this paper are defined on a complete filtered probability space $(\Omega, \mathcal{F}, \mathbb{F} = \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{P})$ satisfying the usual conditions, on which N mutually independent standard scalar Brownian motions $W_1(t), \ldots, W_N(t)$ are defined, one per agent. We write $\mathbb{E}[\cdot]$ and $\mathbb{E}^{\mathcal{F}_t}[\cdot] \triangleq \mathbb{E}[\cdot \mid \mathcal{F}_t]$ for the expectation and the conditional expectation given $\mathcal{F}_t$, respectively.

Deriving from (3) and (4), a strict-feedback representation is adopted for agent i kinematic and dynamic behavior under nonholonomic constraints, with consideration of the uncertainty term $f_{\eta i}$ and disturbance τ_{di}. Beyond this drift-level uncertainty, we account for unmodeled process noise entering the dynamic subsystem – e.g., wheel-torque ripple, unmodeled actuator dynamics, or terrain-induced excitation – as a diffusion term driven by $W_i(t)$, so that the strict-feedback representation is the following Itô stochastic differential equation (SDE):

$$\begin{aligned} dp_i &= g_{pi}(p_i)\,\eta_i\,dt\\ d\eta_i &= \big[f_{\eta i}(p_i, \eta_i) + g_{\eta i}(p_i, \eta_i)\tau_i + k_{\eta i}(p_i, \eta_i)\tau_{di}\big]dt + \sigma_{\eta i}(p_i, \eta_i)\,dW_i(t)\end{aligned} \tag{5}$$

where $\sigma_{\eta i} : \mathbb{R}^3 \times \mathbb{R}^2 \to \mathbb{R}^2$ is the diffusion coefficient of the dynamic subsystem. The kinematic subsystem is left noise-free: p_i is fully determined by the (noise-corrupted) velocity η_i through the exactly known kinematic relation (3), so any process noise entering the robot's motion does so through η_i, not directly through p_i.

Assumption 1. *$f_{\eta i}(\cdot,\cdot)$, $g_{\eta i}(\cdot,\cdot)$, $k_{\eta i}(\cdot,\cdot)$, and $\sigma_{\eta i}(\cdot,\cdot)$ are globally Lipschitz and of at most linear growth in (p_i, η_i), uniformly in t. Consequently, for every admissible control $\tau_i \in \mathcal{L}^2_{\mathbb{F}}(0,\infty;\mathbb{R}^2)$ and every $\mathcal{F}_0$-measurable initial condition with finite second moment, the SDE (5) admits a unique, $\mathbb{F}$-adapted strong solution.*

Similarly, the geometric centroid (x_0, y_0) along with the desired heading θ_0 define the pose of a virtual leader (agent 0). which is assumed to produce a smooth and bounded time-varying trajectory $q_0 = [x_0, y_0, \theta_0]^T \in \mathbb{R}^3$, which satisfies $\dot{x}_0 = v_0\cos(\theta_0), \dot{y}_0 = v_0\sin(\theta_0)$ and $\dot{\theta}_0 = \omega_0$, where $v_0 \in \mathbb{R}$ and $\omega_0 \in \mathbb{R}$ denote desired linear velocity and angular velocity, respectively.

Assumption 2. *The θ_i for $0 \leq i \leq N$ is bounded, ω_i for $0 \leq i \leq N$ is persistently exciting, and $|\omega_i| \leq \omega_{max}$.*

Remark 1. *The angular velocity ω_i is assumed to be persistently exciting, implying that it does not converge to zero. This assumption arises from the nonholonomic nature of the wheeled mobile robot system.*

Assumption 3. *At least one agent in the formation is capable of receiving information from the virtual leader to ensure effective coordination.*

2.3 Problem formulation

The dynamic model of each NWMR is regarded as a strict-feedback nonlinear stochastic system subject to both external disturbances and process noise, formulated as the Itô SDE (5). The target trajectory is generated by the dynamics of an uncontrolled virtual leader, represented as $\dot{p}_0 = g_p(p_0)\,\eta_0$, where $\eta_0 \triangleq [v_0, \omega_0]^T \in \mathbb{R}^2$ denotes the desired velocity. The local neighborhood kinematic and dynamic tracking error $\delta_{pi} \in \mathbb{R}^3$ and $\delta_{\eta i} \in \mathbb{R}^3$, for all $i \in \mathcal{V}$, respectively, are defined as

$$\begin{aligned}\delta_{pi} &\triangleq \sum_{j\in\mathcal{N}_i} a_{ij}(p_i + \Delta_i - p_j - \Delta_j) + b_i(p_i + \Delta_i - p_0),\\ \delta_{\eta i} &\triangleq \sum_{j\in\mathcal{N}_i} a_{ij}(\eta_i - \eta_j) + b_i(\eta_i - \eta_0),\end{aligned} \tag{6}$$

where $\Delta_i \triangleq [d_{i,x}, d_{i,y}]^T \in \mathbb{R}^2$ specifies the target offset between agent i and the leader to maintain the desired formation. Since δ_{pi} is a linear combination of the (noise-free) positions p_i, its temporal derivative retains the deterministic form $\dot{\delta}_{pi} = (b_i + d_i)g_{pi}\eta_i - \sum_{j\in\mathcal{N}_i} a_{ij}g_{pj}\eta_j - b_i g_p(p_0)\eta_0$. Let $\dot{\delta}_p \triangleq [\dot{\delta}_{p1}^T, \ldots, \dot{\delta}_{pn}^T]^T, \dot{\bar{p}}_0 \triangleq (\bar{1}_3 \otimes I_3)\dot{p}_0, p \triangleq [p_1^T, \ldots, p_n^T]^T, g_p \triangleq \mathrm{diag}(g_{p_1}, \ldots, g_{p_n})$, and $\eta \triangleq [\eta_1^T, \ldots, \eta_n^T]^T$, the collective kinematic tracking error vector of the entire multi-agent system is expressed as

$$\dot{\delta}_p = -(\mathcal{B} \otimes I_3)\,\dot{\bar{p}}_0 + \big((\mathcal{L} + \mathcal{B}) \otimes I_3\big) g_p \eta. \tag{7}$$

In contrast, $\delta_{\eta i}$ is a linear combination of the noise-corrupted velocities η_i, η_j, so its stochastic differential inherits a diffusion term from (5): $d\delta_{\eta_i} = \big[f_i(t) + (b_i + d_i)(g_{\eta_i}\tau_i + k_{\eta_i}\tau_{di}) - \sum_{j\in\mathcal{N}_i} a_{ij}(g_{\eta_j}\tau_j + k_{\eta_j}\tau_{dj})\big]dt + (b_i + d_i)\sigma_{\eta i}(p_i, \eta_i)\, dW_i(t) - \sum_{j\in\mathcal{N}_i} a_{ij}\sigma_{\eta j}(p_j, \eta_j)\, dW_j(t)$, where $f_i(t) \triangleq \sum_{j\in\mathcal{N}_i} a_{ij}(f_{\eta_i} - f_{\eta_j}) + b_i(f_{\eta_i} - f_{\eta_0})$. Accordingly, let $\dot{\delta}_\eta \triangleq [\dot{\delta}_{\eta_1}^T, \ldots, \dot{\delta}_{\eta_n}^T]^T, f(t) \triangleq [f_1^T, \ldots, f_n^T]^T, g_\eta \triangleq \mathrm{diag}(g_{\eta_1}, \ldots, g_{\eta_n}), k_\eta \triangleq \mathrm{diag}(k_{\eta_1}, \ldots, k_{\eta_n}), \sigma_\eta \triangleq \mathrm{diag}(\sigma_{\eta 1}, \ldots, \sigma_{\eta n}), \tau \triangleq [\tau_1^T, \ldots, \tau_n^T]^T$, $\tau_d \triangleq [\tau_{d1}^T, \ldots, \tau_{dn}^T]^T$, and $W(t) \triangleq [W_1(t), \ldots, W_n(t)]^T$, the collective dynamic tracking error of all agents now can be expressed as the SDE

$$d\delta_\eta = \Big[f(t) + \big((\mathcal{L} + \mathcal{B}) \otimes I_2\big)\,(g_\eta\tau + k_\eta\tau_d)\Big]dt + \big((\mathcal{L} + \mathcal{B}) \otimes I_2\big)\sigma_\eta(x)\, dW(t). \tag{8}$$

To design distributed optimal control for both dynamics and kinematics, the backstepping approach will be used. Let us define $\eta_a \triangleq [\eta_{a_1}^T, \cdots, \eta_{a_n}^T]^T$ and $\tau_a \triangleq \begin{bmatrix}\tau_{a_1}^T & \cdots & \tau_{a_n}^T\end{bmatrix}^T$ be the feed-forward control inputs designed as

$$\begin{aligned}\eta_a &\triangleq \big(\big((\mathcal{L} + \mathcal{B}) \otimes I_3\big)g_p\big)^{-1}\big((\mathcal{B} \otimes I_3)\dot{\bar{p}}_0 + \big((\mathcal{L} + \mathcal{B}) \otimes I_3\big)g_p e_\eta\big)\,,\\ \tau_a &\triangleq \big(\big((\mathcal{L} + \mathcal{B}) \otimes I_2\big)g_\eta\big)^{-1}\big(-\big((\mathcal{L} + \mathcal{B}) \otimes I_2\big)g_p^T e_p\big)\,.\end{aligned} \tag{9}$$

Let $u_a \triangleq [\eta_a^T, \tau_a^T]^T$ denote auxiliary inputs, and $u^* \triangleq [\eta^{*T}, \tau^{*T}]^T$ denote the distributed optimal feedback control inputs, in which η^* and τ^* will be developed in the subsequent sections. We now define the actual input $u \triangleq \big[\eta^T, \tau^T\big]^T$ as $u \triangleq u^* + u_a$. Let $\delta \triangleq [\delta_1^T, \ldots, \delta_n^T]^T$, where $\delta_i \triangleq [\delta_{pi}^T, \delta_{\eta i}^T]^T$, $x \triangleq [x_1^T, \ldots, x_n^T]^T$, where $x_i \triangleq [p_i^T, \eta_i^T]^T, \vartheta \triangleq [\vartheta_1^T, \ldots, \vartheta_n^T]^T$, where $\vartheta_i \triangleq [\mathbf{0}, \tau_{di}^T]^T, f_e(t) \triangleq [f_{e1}^T, \ldots, f_{en}^T]^T$, where $f_{ei}(t) \triangleq [\mathbf{0}, f_i^T]^T, k(x) \triangleq \mathrm{diag}(k_1, \ldots, k_n)$, where $k_i(x) \triangleq \mathrm{diag}(0, k_{\eta i})$, and $g(x) \triangleq \mathrm{diag}(g_1, \ldots, g_n)$, where $g_i(x) = \mathrm{diag}(g_{pi}, g_{\eta i})$, and $\sigma(x) \triangleq \mathrm{diag}(\sigma_1, \ldots, \sigma_n)$, where $\sigma_i(x_i) \triangleq \mathrm{diag}(0, \sigma_{\eta i})$ places the diffusion coefficient in the same (dynamic) channel as $k_i(x)$ and $g_i(x)$, we now can obtain the SDE governing the variable δ as follows:

$$d\delta = \Big[f_e(t) + \big((\mathcal{L} + \mathcal{B}) \otimes I_5\big)\big(g(x)u^* + k(x)\vartheta\big)\Big]dt + \big((\mathcal{L} + \mathcal{B}) \otimes I_5\big)\sigma(x)\, dW(t). \tag{10}$$

Let us consider the tracking error dynamics of agent i, $i = 1, \ldots, N$ which can be obtained from equation (10):

$$\begin{aligned}d\delta_i = \Big[f_{ei}(t) + (b_i + d_i)\big(g_i(x_i)u_i + k_i(x_i)\vartheta_i\big) - \sum_{j\in\mathcal{N}_i} a_{ij}\big(g_j(x_j)u_j + k_j(x_j)\vartheta_j\big)\Big]dt\\ + (b_i + d_i)\sigma_i(x_i)\, dW_i(t) - \sum_{j\in\mathcal{N}_i} a_{ij}\sigma_j(x_j)\, dW_j(t)\end{aligned} \tag{11}$$

For later use in deriving the stochastic HJI equation (Section 3), it is convenient to collect the $1 + |\mathcal{N}_i|$ mutually independent Brownian motions driving (11) into the vector $\mathcal{W}_i(t) \triangleq \big[W_i(t), \{W_j(t)\}_{j\in\mathcal{N}_i}\big]^T$ and the corresponding diffusion coefficient into the matrix

$$\Sigma_i(x) \triangleq \Big[(b_i + d_i)\sigma_i(x_i),\ \{-a_{ij}\sigma_j(x_j)\}_{j\in\mathcal{N}_i}\Big], \tag{12}$$

so that (11) is written compactly as $d\delta_i = F_i(x,u,\vartheta)\,dt + \Sigma_i(x)\,d\mathcal{W}_i(t)$, with $F_i(x,u,\vartheta)$ denoting the drift term in the brackets of (11).

Lemma 1. *The distributed H_∞ optimal formation tracking problem, in the mean-square sense, for multi-agent systems governed by the stochastic dynamics in equation (5) is equivalent to solving the distributed H_∞ optimal control problem for the system given in equation (10).*

3 Bounded L_2-gain optimal synchronization problem

Suppose the output measurements are given by $y_i = h_i(e_i) \in \mathbb{R}^{n_i}$, where $h_i(e_i)$ is a continuously differentiable and left-invertible mapping. This ensures that $h_i(e_i)$ is directly accessible. Under nonzero disturbances, i.e., $\vartheta_i \neq 0$ and $\vartheta_j \neq 0$, we define the generalized disturbance vector as $\varpi_i \triangleq \begin{bmatrix}\vartheta_i^T & \vartheta_{-i}^T\end{bmatrix}^T$, where $\vartheta_{-i} \triangleq \{\vartheta_j, j \in \mathcal{N}_i\}$. The performance output vector is defined as $\varphi_i \triangleq \begin{bmatrix}\delta_i^T(t) & u_i^T(t) & u_{-i}^T(t)\end{bmatrix}^T$, where $u_{-i} \triangleq \{u_j, j \in \mathcal{N}_i\}$. Since δ_i now evolves according to the Itô SDE (11), a pathwise integral cost is no longer well-defined as a deterministic quantity; the local performance indices are accordingly defined as an expectation over sample paths,

$$J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) \triangleq \mathbb{E}\left[\int_0^\infty r_i(\delta_i(\rho), u_i(\rho), u_{-i}(\rho), \vartheta_i(\rho), \vartheta_{-i}(\rho))d\rho\right], \tag{13}$$

where $r_i : \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}_{\geq 0}$ denotes the instantaneous cost defined as $r_i \triangleq Q_{ii}(e_i) + u_i^T R_{ii} u_i + \sum_{j\in\mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii} \vartheta_i - \gamma^2 \sum_{j\in\mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j$, with $Q_{ii} : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$ being continuously differentiable function satisfying $\underline{q}\|x\|^2 \leq Q_{ii}(x) \leq \overline{q}\|x\|^2$ for $\underline{q}, \overline{q} \in \mathbb{R}_{>0}$, R_{ii} and R_{ij} are positive definite matrices, and γ represents the amount of attenuation from the disturbance input $\vartheta(t)$ to the defined performance output variable $\varphi(t)$.

Definition 1. *The multi-agent control protocols u_i, $i = 1, \ldots, N$, for (5) is said to be admissible on the set Ω denoted by $u_i \in \Psi(\Omega)$, if u_i is continuous with $u_i(0) = 0$, u_i stabilizes (5) on Ω* in the mean-square sense *(i.e., $\mathbb{E}\|\delta_i(t)\|^2 \to 0$ as $t \to \infty$ for every $\mathcal{F}_0$-measurable $\delta_i(0)$ with finite second moment), and $J_i(e_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$ is finite.*

Definition 2. *(Bounded L2-Gain or Disturbance Attenuation)[[15]], [[23]], [[24]] The objective is to identify admissible control inputs that bound the local errors in (7) and (8) by L_2-bounded in the mean-square sense, minimize the cost functional in (13), and meet the disturbance attenuation condition stated below:*

$$\mathbb{E}\left[\int_0^T \|z_i\|^2 dt\right] \leq \gamma^2\, \mathbb{E}\left[\int_0^T \|\omega_i\|^2 dt\right] + \beta_i(e_i(0)), \tag{14}$$

where $z_i \triangleq Q_{ii}(e_i) + u_i^T R_{ii} u_i + \sum_{j\in\mathcal{N}_i} u_j^T R_{ij} u_j$, and $\omega_i \triangleq \vartheta_i^T T_{ii} \vartheta_i + \sum_{j\in\mathcal{N}_i} \vartheta_j^T T_{ij} \vartheta_j$. This is the direct stochastic analogue of the mean-square stabilizability condition used to pose the linear stochastic H_∞ problem in [21], specialized to our nonlinear, agent-coupled setting.

Remark 2. *The disturbance attenuation inequality given in (14) indicates that the effect of external disturbances on the system's expected output performance is constrained within a specified bound determined by the attenuation constant γ. The minimal value of γ that satisfies this condition is known as the optimal robust control gain. Nevertheless, due to the complexity of general nonlinear stochastic systems, there is no closed-form solution to compute this optimal value. As a result, a conservative (i.e., sufficiently large) value of γ is usually selected beforehand to ensure that the control design remains realizable.*

According to [24], the previously described problem can be reformulated as the following zero-sum differential game $J_i(\delta_i(0)) = \min_{u_i} \max_{\vartheta_i} J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$, where J_i is defined in (13). In this setting, the control player minimizes the cost functional J_i, whereas the disturbance player maximizes it. The problem defines a two-player zero-sum differential game, admitting a unique solution if a game-theoretic saddle point (u_i^*, v_i^*) exists such that

$$\begin{aligned} J_i^*(\delta_i(0)) &= \min_{u_i}\max_{\vartheta_i} J_i(\delta_i(0), u_i, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*) \\ &= \max_{\vartheta_i}\min_{u_i} J_i(\delta_i(0), u_i, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*), \end{aligned} \tag{15}$$

where J_i^* denotes the Nash equilibrium of the zero-sum differential graphical game. The corresponding performance satisfies

$$\begin{aligned} J_i(u_i^*, u_{-i}^*, \vartheta_i, \vartheta_{-i}^*) &\le J_i(u_i^*, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*) \\ &\le J_i(u_i, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*). \end{aligned} \tag{16}$$

The distributed local value function for agent i is defined as the conditional expectation

$$V_i\big(\delta_i(t), u_i, u_{-i}, \vartheta_i, \vartheta_{-i}\big) = \mathbb{E}^{\mathcal{F}_t}\left[\int_t^{\infty} r_i\big(\delta_i(\rho), u_i(\rho), u_{-i}(\rho), \vartheta_i(\rho), \vartheta_{-i}(\rho)\big)d\rho\right], \tag{17}$$

which is well-defined by the tower property since δ_i is $\mathbb{F}$-adapted (Assumption 1). Because δ_i now follows the Itô SDE (11) rather than an ordinary differential equation, Leibniz's rule no longer applies to differentiate V_i along trajectories; instead we apply Itô's formula to $V_i(\delta_i(t))$ and invoke Dynkin's formula on the resulting martingale representation. This introduces the infinitesimal generator of (11),

$$\begin{aligned} \mathcal{A}_i V_i \triangleq \nabla V_i^T &\Big(f_{ei} + (b_i + d_i)\big(g_i(x_i)u_i + k_i(x_i)\vartheta_i\big) - \sum_{j\in\mathcal{N}_i} a_{ij}\big(g_j(x_j)u_j + k_j(x_j)\vartheta_j\big)\Big) \\ &+ \frac{1}{2}\mathrm{tr}\Big(\Sigma_i(x)^T\nabla^2 V_i(\delta_i)\,\Sigma_i(x)\Big), \end{aligned} \tag{18}$$

where $\nabla V_i \triangleq \frac{\partial V_i}{\partial \delta_i} \in \mathbb{R}^n$ is the gradient, $\nabla^2 V_i \triangleq \frac{\partial^2 V_i}{\partial\delta_i\partial\delta_i^T} \in \mathbb{R}^{n\times n}$ is the Hessian of V_i with respect to δ_i, and $\Sigma_i(x)$ is the stacked diffusion coefficient of (11) defined in (12). The trace term $\frac{1}{2}\mathrm{tr}(\Sigma_i^T\nabla^2 V_i\,\Sigma_i)$ is the genuine stochastic (Itô) correction with no deterministic counterpart; setting $\Sigma_i \equiv 0$ collapses $\mathcal{A}_i$ back to the ordinary drift-only directional derivative used in the deterministic formulation. The Hamilton–Jacobi–Isaacs (HJI) equation, also known as the Bellman equation, is then

$$H_i\Big(\delta_i, \nabla V_i, \nabla^2 V_i, u_i, u_{-i}, \vartheta_i, \vartheta_{-i}\Big) \triangleq \mathcal{A}_i V_i + r_i(\delta_i, u_i, u_{-i}, \vartheta_i, \vartheta_{-i}) = 0. \tag{19}$$

Since $\Sigma_i(x)$ does not depend on u_i or ϑ_i under the process-noise model of (5) (the diffusion enters additively rather than through the control channel), the trace term in $\mathcal{A}_i V_i$ is unaffected by the stationary conditions below; the optimal control signals u_i^* and the optimal disturbance signals ϑ_i^* are obtained exactly as in the deterministic case, from $\frac{\partial H_i}{\partial u_i} = 0$ and $\frac{\partial H_i}{\partial \vartheta_i} = 0$, respectively, as follows

$$u_i^*(\delta_i) = -\frac{1}{2}(b_i + d_i)R_{ii}^{-1}g_i^T(x_i)\nabla V_i^*, \tag{20}$$

$$\vartheta_i^*(\delta_i) = \frac{1}{2\gamma^2}(b_i + d_i)T_{ii}^{-1}k_i^T(x_i)\nabla V_i^*. \tag{21}$$

Based on the substitution of the optimal control and disturbance strategies from (20) and (21) into (19), one can derive the associated coupled stochastic Hamilton–Jacobi–Isaacs (HJI) equations governing the optimal value functions.

$$\begin{aligned} &Q_{ii}(\delta_i) + \nabla V_i F_i + \frac{1}{4}(b_i + d_i)^2\nabla V_i^T g_i R_{ii}^{-1} g_i^T \nabla V_i \\ &+ \frac{1}{4}\sum_{j\in\mathcal{N}_i}(b_j + d_j)^2\nabla V_j^T g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \nabla V_j \\ &- \frac{1}{4\gamma^2}\sum_{j\in\mathcal{N}_i}(b_j + d_j)^2\nabla V_j^T k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \nabla V_j \\ &- \frac{1}{4\gamma^2}(b_i + d_i)^2\nabla V_i^T k_i T_{ii}^{-1} k_i^T \nabla V_i \\ &+ \frac{1}{2}\mathrm{tr}\Big(\Sigma_i(x)^T\nabla^2 V_i(\delta_i)\,\Sigma_i(x)\Big) = 0. \end{aligned} \tag{22}$$

where $F_i \triangleq f_{ei}(t) - \frac{1}{2}(b_i + d_i)^2 g_i R_{ii}^{-1} g_i^T \nabla V_i + \frac{1}{2}\sum_{j\in\mathcal{N}_i} a_{ij}(b_j + d_j) g_j R_{jj}^{-1} g_j^T \nabla V_j + \frac{1}{2\gamma^2}(b_i + d_i)^2 k_i T_{ii}^{-1} k_i^T \nabla V_i - \frac{1}{2\gamma^2}\sum_{j\in\mathcal{N}_i} a_{ij}(b_j + d_j) k_j T_{jj}^{-1} k_j^T \nabla V_j$, exactly as in the deterministic case, since the trace term does not involve u_i or ϑ_i and therefore does not enter F_i. The added trace term is the nonlinear, agent-coupled analogue of the extra $C^T PC$ and $D^T PD$ diffusion terms that distinguish the Generalized Algebraic Riccati Equation (GARE) of [21] from the deterministic Riccati equation: there, $V(x) = x^T Px$ and $\Sigma(x) = Cx + Du$ are both quadratic/linear-in-state, so $\frac{1}{2}\mathrm{tr}(\Sigma^T \nabla^2 V\, \Sigma)$ reduces exactly to the algebraic terms $x^T(C^T PC)x$ and $u^T(D^T PD)u$ appearing in the GARE; here V_i and Σ_i are general nonlinear functions of δ_i, so the correction survives only as a trace of a state-dependent Hessian rather than as a constant coefficient matrix, and admits no closed-form (algebraic) solution – which is precisely why the reinforcement-learning approach developed in the remainder of this paper is needed. The closed-loop system corresponds to agent i. For a given solution V_i, define $u_i^* = u_i(V_i)$, $u_{-i}^* = u_{-i}(V_i)$, $\vartheta_i^* = \vartheta_i(V_i)$, and $\vartheta_{-i}^* = \vartheta_{-i}(V_i)$ in terms of V_i, then (22) can be written as

$$H_i\left(e_i, \nabla V_i, \nabla^2 V_i, u_i^*, u_{-i}^*, \vartheta_i^*, \vartheta_{-i}^*\right) = 0, \tag{23}$$

with $V_i(0) = 0$. The optimal tracking control problem for multiple nonholonomic WMRs formulated as a stochastic zero-sum game leads to the challenge of solving a set of coupled stochastic HJI equations corresponding to the nonlinear dynamics in (11). The intrinsic nonlinearity of these systems renders an analytical solution intractable regardless of whether the dynamics are deterministic or stochastic, and the added trace term rules out the closed-form (GARE-based) route available in the linear stochastic case. Consequently, the remainder of this paper is devoted to addressing this issue via a reinforcement learning-based approach.

4 Online integral reinforcement learning algorithm for multi-agent cooperative games

4.1 IRL and value function approximation

Leveraging the IRL concept proposed in [ref], we aim to reformulate the Bellman equation in a way that bypasses explicit use of system dynamics. Since δ_i is now an Itô process governed by (11) and V_i is defined as the conditional expectation (17), the deterministic IRL identity no longer holds pathwise; instead, applying Itô's formula to $V_i(\delta_i(\tau))$ on $[t-T, t]$, using $\mathcal{A}_i V_i + r_i = 0$ (19), and taking the conditional expectation $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$ to annihilate the resulting local-martingale term (the standard Dynkin's-formula argument, mirroring [21, Eq. (ito_2)]) gives, for any time t over a finite interval $[t-T, t]$ of length $T > 0$,

$$\begin{aligned} V_i(\delta_i(t-T)) = \mathbb{E}^{\mathcal{F}_{t-T}}\Bigg[\int_{t-T}^{t} \Big(Q_{ii}(\delta_i) + u_i^T R_{ii} u_i + \sum_{j\in\mathcal{N}_i} u_j^T R_{ij} u_j \\ - \gamma^2 \vartheta_i^T T_{ii}\vartheta_i - \gamma^2 \sum_{j\in\mathcal{N}_i} \vartheta_j^T T_{ij}\vartheta_j\Big) d\tau + V_i(\delta_i(t))\Bigg]. \end{aligned} \tag{24}$$

Notably, the IRL-based representation of the Bellman equation eliminates the explicit dependency on system dynamics, exactly as in the deterministic case; the only change is that (24) holds in conditional expectation rather than pathwise, since a single realized trajectory carries an extra zero-conditional-mean fluctuation contributed by the diffusion term of (11) that only cancels on average. To facilitate solving the value function $V_i(\delta_i)$ in the IRL-based Bellman equation (24), we consider approximating it using function approximation techniques. Based on the Weierstrass higher-order approximation theorem [], for each $i \in N$, both the value function V_i and its gradient ∇V_i can be approximated with arbitrary accuracy as follows:

$$V_i(\delta_i) = W_i^T \phi_i(\delta_i) + \epsilon_i \tag{25}$$

$$\nabla V_i(\delta_i) = \nabla \phi_i^T(\delta_i) W_i + \nabla \epsilon_i \tag{26}$$

Where $\phi_i(\delta_i) \in \mathbb{R}^L$ is the critic NNs activation function vectors, $W_i \in \mathbb{R}^L$ are the ideal weight parameter vectors, with L denotes the number of neurons in the critic NNs hidden layer. $\epsilon_i(\delta_i)$ is the NN approximation error. The gradients are given by $\nabla\phi_i(\delta_i) = \dfrac{\partial \phi_i}{\partial \delta_i}$ and $\nabla\epsilon_i(\delta_i) = \dfrac{\partial \epsilon_i}{\partial \delta_i}$. It is worth mentioning that $\epsilon_i(\delta_i)$ converges to zero uniformly as $L \to \infty$.

Assumption 4. *$\phi_i(\delta_i), \nabla\phi_i(\delta_i), \nabla^2\phi_i(\delta_i), \epsilon_i, \nabla\epsilon_i$, $\nabla^2\epsilon_i$, and the diffusion coefficient $\Sigma_i(x)$ of (11) are bounded on the compact set Ω such that $\|\phi_i\| \leq b_{\phi_i}, \|\nabla\phi_i\| \leq b_{\nabla\phi_i}, \|\nabla^2\phi_i\| \leq b_{\nabla^2\phi_i}, \|\epsilon_i\| \leq b_{\epsilon_i}, \|\nabla\epsilon_i\| \leq b_{\nabla\epsilon_i}, \|\nabla^2\epsilon_i\| \leq b_{\nabla^2\epsilon_i}$, and $\|\Sigma_i(x)\| \leq b_{\Sigma_i}$.*

Then, the Bellman equation (24) can be approximated at each step as

$$\mathbb{E}^{\mathcal{F}_{t-T}}\left[\int\limits_{t-T}^{t} \Big(Q_{ii}(\delta_i) + u_i^T R_{ii} u_i + \sum_{j\in\mathcal{N}_i} u_j^T R_{ij} u_j - \gamma^2 \vartheta_i^T T_{ii}\vartheta_i - \gamma^2 \sum_{j\in\mathcal{N}_i} \vartheta_j^T T_{ij}\vartheta_j\Big) d\tau\right] = -W_i^T\, \mathbb{E}^{\mathcal{F}_{t-T}}\big[\Delta\phi_i\left(\delta_i(t)\right)\big] + \bar{\epsilon}_{B_i} \quad (27)$$

where $\Delta\phi_i(\delta_i(t)) \triangleq \phi_i(\delta_i(t)) - \phi_i(\delta_i(t-T))$ and $\bar{\epsilon}_{B_i} \triangleq -\mathbb{E}^{\mathcal{F}_{t-T}}[\epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t-T))]$. Since $\delta_i(\tau)$ is now an Itô process, $\Delta\phi_i(\delta_i(t))$ and $\epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t-T))$ must be expanded by Itô's formula rather than the ordinary chain rule: writing $F_i(\tau) \triangleq f_{ei} + (b_i + d_i)(g_i u_i + k_i\vartheta_i) - \sum_{j\in\mathcal{N}_i} a_{ij}(g_j u_j + k_j\vartheta_j)$ for the drift of (11),

$$\Delta\phi_i(\delta_i(t)) = \int\limits_{t-T}^{t} \Big[\nabla\phi_i^T F_i + \tfrac{1}{2}\mathrm{tr}\big(\Sigma_i^T \nabla^2\phi_i\, \Sigma_i\big)\Big] d\tau + \int\limits_{t-T}^{t} \nabla\phi_i^T \Sigma_i\, d\mathcal{W}_i(\tau),$$

$$\epsilon_i(\delta_i(t)) - \epsilon_i(\delta_i(t-T)) = \int\limits_{t-T}^{t} \Big[\nabla\epsilon_i^T F_i + \tfrac{1}{2}\mathrm{tr}\big(\Sigma_i^T \nabla^2\epsilon_i\, \Sigma_i\big)\Big] d\tau + \int\limits_{t-T}^{t} \nabla\epsilon_i^T \Sigma_i\, d\mathcal{W}_i(\tau), \quad (28)$$

each with a drift-plus-Itô-correction integral (deterministic, given the trajectory) and a local-martingale term that is $\mathcal{F}_{t-T}$-conditionally mean-zero. Taking $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$ removes both martingale terms from (27) itself, leaving $\bar{\epsilon}_{B_i} = -\int\limits_{t-T}^{t} \mathbb{E}^{\mathcal{F}_{t-T}}\Big[\nabla\epsilon_i^T F_i + \frac{1}{2}\mathrm{tr}(\Sigma_i^T \nabla^2\epsilon_i\, \Sigma_i)\Big] d\tau$, which reduces exactly to the deterministic ϵ_{B_i} of the noise-free case when $\Sigma_i \equiv 0$. In the online, single-trajectory implementation used by the adaptive law below, the realized (non-averaged) residual is used directly rather than its conditional expectation; write

$$\epsilon_{B_i}(t) \triangleq \bar{\epsilon}_{B_i} - \int\limits_{t-T}^{t} \nabla\epsilon_i^T \Sigma_i\, d\mathcal{W}_i(\tau) \quad (29)$$

for the realized value, so that $\mathbb{E}^{\mathcal{F}_{t-T}}[\epsilon_{B_i}(t)] = \bar{\epsilon}_{B_i}$: using $\epsilon_{B_i}(t)$ in place of $\bar{\epsilon}_{B_i}$ in (27) and in the adaptive law of Section 4.2 introduces an additional, $\mathcal{F}_{t-T}$-conditionally-mean-zero perturbation relative to the noise-free case – it does not bias the residual in expectation, but it does increase its variance, which is why Theorem 1 below concludes boundedness in the mean-square sense rather than pathwise. Under Assumption 4, $\epsilon_{B_i}(t)$ is bounded in mean square on the compact set Ω: by the triangle inequality on the drift and Itô-correction integrals and the Itô isometry on the martingale term (which gives $\mathbb{E}^{\mathcal{F}_{t-T}}\big\| \int_{t-T}^{t} \nabla\epsilon_i^T \Sigma_i\, d\mathcal{W}_i \big\|^2 = \int_{t-T}^{t} \mathbb{E}^{\mathcal{F}_{t-T}}\|\nabla\epsilon_i^T \Sigma_i\|^2 d\tau \leq T\, b_{\nabla\epsilon_i}^2 b_{\Sigma_i}^2$), there exists a bound ϵ_{max} such that $\mathbb{E}^{\mathcal{F}_{t-T}}\|\epsilon_{B_i}(t)\|^2 \leq \epsilon_{max}^2$ (the deterministic case, $\|\epsilon_{B_i}\| \leq \epsilon_{max}$ pathwise, per [17], is recovered exactly when $\Sigma_i \equiv 0$). Analogously to (28), $\Delta\phi_i(\delta_i(t))$ used operationally in the adaptive law below is its realized (pathwise) value, i.e. the full right-hand side of the first line of (28), not its conditional expectation.

4.2 Critic network weight convergence and tuning via experience replay

An online integral reinforcement learning (IRL) approach is introduced, in which a single neural network—the critic—is employed to approximate both the value function and the optimal policy derived from the IRL-based Bellman formulation. Unlike traditional methods, this framework enables the reuse of collected experience data, leading to a practical and verifiable condition for the convergence of the critic's weight parameters. Since the true ideal weight vectors W_i are not available, we utilize current estimates $\hat{W}_i$ during the learning process. Based on these estimates, the value function V_i and its gradient ∇V_i can be approximated accordingly.

$$\hat{V}_i(\delta_i) = \hat{W}_i^T \phi_i(\delta_i) \tag{30}$$

$$\nabla \hat{V}_i(\delta_i) = \nabla \phi_i^T(\delta_i) \hat{W}_i \tag{31}$$

Consequently, the H_∞ optimal control policy and the corresponding worst-case disturbance strategy can be expressed in terms of the approximated value function and its gradient as follows:

$$\hat{u}_i = -\frac{1}{2}(b_i + d_i) R_{ii}^{-1} g_i^T \nabla\phi_i(\delta_i)^T \hat{W}_i, \tag{32}$$

$$\hat{\vartheta}_i = \frac{1}{2\gamma^2}(b_i + d_i) T_{ii}^{-1} k_i^T \nabla\phi_i(\delta_i)^T \hat{W}_i, \tag{33}$$

Using the value-function approximation (30) together with the IRL-based Bellman equation (27), the approximate Bellman equation becomes $e_i(t) = \hat{W}_i^T \Delta\phi_i(\delta_i(t)) + \varrho_i(t)$, where $\varrho_i(t) \triangleq \int_{t-T}^{t} \left(Q_{ii}(\delta_i) + \hat{u}_i^T R_{ii} \hat{u}_i + \sum_{j\in\mathcal{N}_i} \hat{u}_j^T R_{ij} \hat{u}_j - \gamma^2 \hat{\vartheta}_i^T T_{ii} \hat{\vartheta}_i - \gamma^2 \sum_{j\in\mathcal{N}_i} \hat{\vartheta}_j^T T_{ij} \hat{\vartheta}_j \right) d\tau$. Substituting (30)-(33) into (22), we can obtain

$$\begin{aligned}
&\hat{H}_i(\delta_i, \hat{W}_i, \hat{u}_i, \hat{u}_{-i}, \hat{\vartheta}_i, \hat{\vartheta}_{-i}) \\
&= Q_{ii}(\delta_i) + \hat{W}_i^T \nabla\phi_i(\delta_i) f_{ei}(t) \\
&\quad - \frac{1}{4}(b_i + d_i)^2 \hat{W}_i^T \nabla\phi_i(\delta_i) g_i R_{ii}^{-1} g_i^T \nabla\phi_i(\delta_i)^T \hat{W}_i \\
&\quad + \frac{1}{4\gamma^2}(b_i + d_i)^2 \hat{W}_i^T \nabla\phi_i(\delta_i) k_i T_{ii}^{-1} k_i^T \nabla\phi_i(\delta_i)^T \hat{W}_i \\
&\quad + \frac{1}{4} \sum_{j\in\mathcal{N}_i} (b_j + d_j)^2 \hat{W}_j^T \nabla\phi_j(\delta_j) g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \nabla\phi_j(\delta_j)^T \hat{W}_j \\
&\quad - \frac{1}{4\gamma^2} \sum_{j\in\mathcal{N}_i} (b_j + d_j)^2 \hat{W}_j^T \nabla\phi_j(\delta_j) k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \nabla\phi_j(\delta_j)^T \hat{W}_j \\
&\quad + \frac{1}{2} \hat{W}_i^T \nabla\phi_i(\delta_i) \sum_{j\in\mathcal{N}_i} a_{ij}(b_j + d_j) g_j R_{jj}^{-1} g_j^T \nabla\phi_j(\delta_j)^T \hat{W}_j \\
&\quad - \frac{1}{2\gamma^2} \hat{W}_i^T \nabla\phi_i(\delta_i) \sum_{j\in\mathcal{N}_i} a_{ij}(b_j + d_j) k_j T_{jj}^{-1} k_j^T \nabla\phi_j(\delta_j)^T \hat{W}_j
\end{aligned} \tag{34}$$

The Itô-correction trace term of the stochastic coupled HJI equation (22) is omitted from (34): it depends only on $\Sigma_i(x)$ and $\nabla^2\phi_i(\delta_i)$, not on $\hat{W}_i$, $\hat{u}_i$, or $\hat{\vartheta}_i$, so it contributes no gradient with respect to the weights being adapted and drops out of the weight-update law derived below in exactly the same way it dropped out of the stationary conditions (20)–(21); it is instead accounted for through $\epsilon_{B_i}(t)$ in (29), which is where the IRL-integral route below actually uses it.

To facilitate the convergence of the estimated weight vector $\hat{W}_i$, a concurrent learning approach incorporating experience replay is employed. This method leverages both previously stored historical data and real-time data to simultaneously update $\hat{W}_i$. As in (29), every quantity below ($\Delta\phi_i$, Ψ_i, Θ_i, and ϱ_i) is computed from the single realized trajectory observed online, not from a conditional expectation over sample paths – consistent with treating (35) as a stochastic-approximation recursion driven by noisy (single-path) Bellman residuals rather than a batch regression over multiple rollouts. The following presents the proposed gradient-descent-based learning algorithm for the critic neural network using a novel experience replay mechanism:

$$\dot{\hat{W}}_i = -\alpha_i \frac{\Delta\phi_i(t)}{\left(\Delta\phi_i(t)^T\Delta\phi_i(t)+1\right)^2}\Psi_i(t) + \xi_i(t)\Theta_i(t) + \tag{35}$$

$$\sum_{k=1}^{l}\left(-\alpha_i \frac{\Delta\phi_i(t_k)}{\left(\Delta\phi_i^T(t_k)\Delta\phi_i(t_k)+1\right)^2}\Psi_i(t_k) + \xi_i(t_k)\Theta_i(t_k)\right) \tag{36}$$

where $\Psi_i(t) \triangleq \Delta\phi_i(t)^T\hat{W}_i + \varrho_i(t)$. Stability of the closed-loop system is ensured by the second term in (35), formulated as $\xi_i(t) = 0$, if $\delta_i^T(t)\delta_i(t) \le \delta_i^T(t-T)\delta_i(t-T)$, and $\xi_i(t) = 1$, otherwise; and the term Θ_i is defined as

$$\Theta_i(t) = -\beta_i \int_{t-T}^{t} \left(\frac{1}{2}(b_i+d_i)\nabla\phi_i g_i R_{ii}^{-1} g_i^T \delta_i - \frac{1}{2\gamma^2}(b_i+d_i)\nabla\phi_i k_i T_{ii}^{-1} k_i^T \delta_i\right. \tag{37}$$

$$+ \sum_{j\in\mathcal{N}_i} \frac{1}{2} a_{ij}(b_j+d_j)\nabla\phi_j g_j R_{jj}^{-1} R_{ij} R_{jj}^{-1} g_j^T \delta_j \tag{38}$$

$$\left. - \sum_{j\in\mathcal{N}_i} \frac{1}{2} a_{ij}(b_j+d_j)\nabla\phi_j k_j T_{jj}^{-1} T_{ij} T_{jj}^{-1} k_j^T \delta_j\right) d\tau \tag{39}$$

Unlike $\Psi_i(t)$, the stabilizing term $\Theta_i(t)$ does not depend on $\hat{W}_i$: it is built only from the known regressor gradients $\nabla\phi_i, \nabla\phi_j$, the fixed design matrices g_i, k_i, R_{ii}, T_{ii} (and their neighbor counterparts), and the measured tracking errors δ_i, δ_j. Consequently, under Assumption 4 and boundedness of δ_i, δ_j on the compact set Ω, $\Theta_i(t)$ is a bounded known signal rather than an estimation error: there exists a constant $\theta_{max_i} > 0$, independent of $\hat{W}_i$, such that $\|\Theta_i(t)\| \le \theta_{max_i}$ for all t. This bound is used below to account for the contribution of Θ_i to the weight-error dynamics (43) and to the convergence thresholds of Theorem 1.

The sample data are accumulated in a history stack, which serves as a memory buffer for experience replay. To populate this history stack, data are recorded during the system's online operation, capturing state trajectories and corresponding temporal difference errors under the current policy. Consider $\Delta\phi_i(t_k)$ and $\varrho_i(t_k)$ as evaluated values of $\Delta\phi_i(t)$ and $\varrho_i(t)$ at the recorded time t_k, defined as $\Delta\phi_i(\delta_i(t_k)) \triangleq \phi_i(\delta_i(t_k)) - \phi_i(\delta_i(t_k - T))$.

Condition 1. *Let $Z_i = \{\Delta\bar{\phi}_i^1, \ldots, \Delta\bar{\phi}_i^l\}$ denote the recorded history of regressor data, rank$(Z_i) = L$. This implies that the recorded set contains L linearly independent elements, which matches the number of neurons used in the neural network structure in (25). The history stack is assumed to store a fixed number of samples l, satisfying $l > L$*

Let the estimation error for the adaptive weights $\hat{W}_i$ be defined as $\tilde{W}_i = W_i - \hat{W}_i$. Differentiating both sides yields $\dot{\tilde{W}}_i = -\dot{\hat{W}}_i$. Since (35) also carries the stabilizing term $\xi_i(t)\Theta_i(t) + \sum_{k=1}^{l}\xi_i(t_k)\Theta_i(t_k)$, this term must be carried through with the opposite sign as well. For brevity, collect the regressor Gram matrix and the two bounded perturbation contributions to (35) as

$$\Xi_i \triangleq \Delta\bar{\phi}_i(t)\Delta\bar{\phi}_i(t)^T + \sum_{k=1}^{l}\Delta\bar{\phi}_i(t_k)\Delta\bar{\phi}_i(t_k)^T, \tag{40}$$

$$\mathcal{E}_i(t) \triangleq \frac{\Delta\bar{\phi}_i(t)}{m_i}\epsilon_{B_i}(t) + \sum_{k=1}^{l}\frac{\Delta\bar{\phi}_i(t_k)}{m_i(t_k)}\epsilon_{B_i}(t_k), \tag{41}$$

$$\mathcal{T}_i(t) \triangleq \xi_i(t)\Theta_i(t) + \sum_{k=1}^{l}\xi_i(t_k)\Theta_i(t_k), \tag{42}$$

where $\Delta\bar{\phi}_i(t) = \dfrac{\Delta\phi_i(t)}{\Delta\phi_i(t)^T\Delta\phi_i(t)+1}, m_i = \Delta\phi_i(t)^T\Delta\phi_i(t) + 1, \Delta\bar{\phi}_i(t_k) = \dfrac{\Delta\phi_i(t_k)}{\Delta\phi_i(t_k)^T\Delta\phi_i(t_k)+1}$, and $m_i(t_k) = \Delta\phi_i(t_k)^T\Delta\phi_i(t_k) + 1$. The error dynamics can then be expressed compactly, based on equation (35), as

$$\dot{\tilde{W}}_i(t) = -\alpha_i\Xi_i\,\tilde{W}_i(t) + \alpha_i\,\mathcal{E}_i(t) - \mathcal{T}_i(t). \tag{43}$$

Since $\xi_i \in \{0,1\}$, $\mathbb{E}\|\epsilon_{B_i}(t)\|^2 \leq \epsilon_{max}^2$ (established after (27)), and $\|\Theta_i(t)\| \leq \theta_{max_i}$ (established after (37)), Cauchy–Schwarz gives $\alpha_i\,\mathbb{E}[\tilde{W}_i^T\mathcal{E}_i(t)] \leq \alpha_i(l+1)\epsilon_{max}\,\mathbb{E}\|\tilde{W}_i\|$ and $-\mathbb{E}[\tilde{W}_i^T\mathcal{T}_i(t)] \leq (l+1)\theta_{max_i}\,\mathbb{E}\|\tilde{W}_i\|$ (with at most l terms whenever $\xi_i(t) = 0$, since $\mathcal{T}_i(t)$ then reduces to the historical sum alone); these two bounds are used throughout the proof below.

The complete procedure is summarized in Algorithm 1. Note that steps 3–7 are the only ones that evaluate the input matrices g_i, k_i; the drift f_{ei} and the diffusion Σ_i are never needed, the former because the IRL representation (24) eliminates it and the latter because it is absorbed into the generator identity. Section 4.3 removes the remaining dependence on g_i, k_i as well.

Algorithm 1 Model-Based Stochastic IRL for the H_∞ Graphical Game

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Require: $Q_{ii} \succeq 0$, $R_{ij}, T_{ij} \succ 0$, $\gamma > 0$, $\alpha_i, \beta_i > 0$, $T > 0$; critic basis $\phi_i(\delta_i) \in \mathbb{R}^L$; admissible $\hat{W}_i(0)$; stack size $l > L$. Known: g_i, k_i.

1: **loop** for each t

2: Measure $\delta_i(t)$; receive $\{\delta_j, \hat{W}_j\}_{j\in\mathcal{N}_i}$.

3: $\hat{u}_i \leftarrow -\frac{1}{2}(b_i+d_i)R_{ii}^{-1}g_i^T\nabla\phi_i^T\hat{W}_i,\quad \hat{\vartheta}_i \leftarrow \frac{1}{2\gamma^2}(b_i+d_i)T_{ii}^{-1}k_i^T\nabla\phi_i^T\hat{W}_i$

4: $\Delta\phi_i(t) \leftarrow \phi_i(\delta_i(t)) - \phi_i(\delta_i(t-T)),\quad \varrho_i(t) \leftarrow \int_{t-T}^{t} r_i\,d\tau$

5: $\Psi_i(t) \leftarrow \Delta\phi_i(t)^T\hat{W}_i + \varrho_i(t)$

6: $\xi_i(t) \leftarrow \mathbb{1}\{\delta_i^T(t)\delta_i(t) > \delta_i^T(t-T)\delta_i(t-T)\}$

7: $\Theta_i(t) \leftarrow$ (37) ▷ requires g_i, k_i, g_j, k_j

8: $Z_i \leftarrow Z_i \cup \{\Delta\bar{\phi}_i(t)\}$ while $\mathrm{rank}(Z_i) < L$ ▷ Condition 1

9: $\dot{\hat{W}}_i \leftarrow -\alpha_i\dfrac{\Delta\phi_i(t)\Psi_i(t)}{\left(\Delta\phi_i^T(t)\Delta\phi_i(t)+1\right)^2} + \xi_i(t)\Theta_i(t) + \displaystyle\sum_{k=1}^{l}\left(-\alpha_i\frac{\Delta\phi_i(t_k)\Psi_i(t_k)}{\left(\Delta\phi_i^T(t_k)\Delta\phi_i(t_k)+1\right)^2} + \xi_i(t_k)\Theta_i(t_k)\right)$

10: **end loop**

Ensure: $\hat{W}_i \to W_i,\ \hat{V}_i = \hat{W}_i^T\phi_i,\ \hat{u}_i, \hat{\vartheta}_i$ mean-square UUB (Theorem 1).

4.3 Model-free extension via off-policy integral reinforcement learning

The formulation developed so far is already model-free *in the drift*: the IRL representation (24) removes f_{ei} from the Bellman equation entirely, and the diffusion Σ_i never has to be evaluated – it is absorbed into the generator identity $\mathcal{A}_iV_i + r_i = 0$ and enters the analysis only through the residual bound ϵ_{max} and the constant c_{σ_i}. What is *not* yet model-free is the input channel: the policies (32)–(33) and the stabilizing term (37) evaluate $g_i(x_i)$ and $k_i(x_i)$ explicitly. This subsection removes that last requirement, so that no component of (11) – neither f_{ei}, nor g_i, k_i, nor Σ_i – need be known. The construction is the nonlinear, agent-coupled counterpart of the off-policy reformulation that makes policy iteration model-free for the *linear* stochastic zero-sum game in [21]; there, the unknown matrices A, B_1, B_2, C, D survive only inside the lumped products $B_1^TP + D^TPC$, B_2^TP and D^TPD, which are then identified jointly with P by least squares. The same mechanism is available here, with the ideal weights W_i playing the role of P and the stationarity conditions (20)–(21) supplying the lumping.

Behaviour–target separation. Let u_j, ϑ_j denote the signals *actually* applied to and acting on the plant (the behaviour data, generated as $u_j = \hat{u}_j + e_j$ with a bounded exploration signal e_j), and let $\hat{u}_j, \hat{\vartheta}_j$ denote the current *target* policies (32)–(33) being evaluated and improved. Adding and subtracting the target policies in (11) gives the off-policy form

$$d\delta_i = \hat{F}_i \, dt + \Delta_i \, dt + \Sigma_i(x) \, d\mathcal{W}_i(t), \tag{44}$$

where $\hat{F}_i \triangleq f_{ei} + (b_i + d_i)(g_i \hat{u}_i + k_i \hat{\vartheta}_i) - \sum_{j \in \mathcal{N}_i} a_{ij}(g_j \hat{u}_j + k_j \hat{\vartheta}_j)$ is the drift under the target policies and

$$\Delta_i \triangleq (b_i + d_i)\big[g_i(u_i - \hat{u}_i) + k_i(\vartheta_i - \hat{\vartheta}_i)\big] - \sum_{j \in \mathcal{N}_i} a_{ij}\big[g_j(u_j - \hat{u}_j) + k_j(\vartheta_j - \hat{\vartheta}_j)\big] \tag{45}$$

collects the behaviour–target mismatch. Only $\hat{F}_i$ carries the unknown drift f_{ei}, and it is exactly $\hat{F}_i$ that the Bellman identity eliminates.

Model-free lumping of the input channels. Applying Itô's formula to $V_i(\delta_i)$ along (44), using $\nabla V_i^T \hat{F}_i + \frac{1}{2}\mathrm{tr}(\Sigma_i^T \nabla^2 V_i \Sigma_i) = \mathcal{A}_i V_i = -r_i$ from (19) evaluated at the target policies, and taking $\mathbb{E}^{\mathcal{F}_{t-T}}[\cdot]$ to annihilate the martingale exactly as in (24), leaves only $\nabla V_i^T \Delta_i$ to be dealt with. The decisive observation is that the stationarity conditions (20)–(21) can be read *backwards*: rather than producing the policy from ∇V_i, they express the unknown products $\nabla V_i^T g_i$ and $\nabla V_i^T k_i$ in terms of the policy itself,

$$(b_i + d_i) \, \nabla V_i^T g_i(x_i) = -2 \, \hat{u}_i^T R_{ii}, \qquad (b_i + d_i) \, \nabla V_i^T k_i(x_i) = 2\gamma^2 \, \hat{\vartheta}_i^T T_{ii}, \tag{46}$$

in which R_{ii} and T_{ii} are design weights and $\hat{u}_i, \hat{\vartheta}_i$ are the signals the agent itself generates. Both sides of (46) are therefore computable online without g_i or k_i. This is precisely the step by which $2x^\top PB(\cdot)$ becomes $2x^\top \hat{K}^\top R(\cdot)$ in [21]. The neighbour terms of (45) admit no such identity, since no stationarity condition relates agent i's gradient ∇V_i to agent j's input matrix g_j; they are instead retained as unknown *cross-channel* maps

$$\Xi_{ij}^u(\delta) \triangleq a_{ij} \, g_j^T(x_j) \nabla V_i(\delta_i) \in \mathbb{R}^{m_j}, \qquad \Xi_{ij}^\vartheta(\delta) \triangleq a_{ij} \, k_j^T(x_j) \nabla V_i(\delta_i) \in \mathbb{R}^{q_j}, \tag{47}$$

to be identified from data alongside W_i. Substituting (46)–(47) into $\nabla V_i^T \Delta_i$ and using the NN representation (25) yields the *off-policy stochastic IRL Bellman equation*

$$\begin{aligned} W_i^T \Delta\phi_i(\delta_i(t)) + \mathbb{E}^{\mathcal{F}_{t-T}}\Bigg[\int_{t-T}^{t} \Big(2\hat{u}_i^T R_{ii}(u_i - \hat{u}_i) - 2\gamma^2 \hat{\vartheta}_i^T T_{ii}(\vartheta_i - \hat{\vartheta}_i) \\ + \sum_{j \in \mathcal{N}_i} \big(\Xi_{ij}^{uT}(u_j - \hat{u}_j) + \Xi_{ij}^{\vartheta T}(\vartheta_j - \hat{\vartheta}_j)\big)\Big) d\tau\Bigg] = -\,\mathbb{E}^{\mathcal{F}_{t-T}}\Bigg[\int_{t-T}^{t} r_i \, d\tau\Bigg] + \bar{\epsilon}_{B_i}, \end{aligned} \tag{48}$$

in which every quantity other than the unknowns $W_i, \Xi_{ij}^u, \Xi_{ij}^\vartheta$ is either measured (δ_i, u_j, ϑ_j), self-generated ($\hat{u}_i, \hat{\vartheta}_i$), communicated by neighbours ($\hat{u}_j, \hat{\vartheta}_j$, already required by ϱ_i in the model-based case), or a design choice ($Q_{ii}, R_{ij}, T_{ij}, \gamma$). Setting $u_j \equiv \hat{u}_j$ and $\vartheta_j \equiv \hat{\vartheta}_j$ (no exploration, on-policy data) collapses (48) back to (27) exactly.

Actor parameterisation and the enlarged regressor. Because the policies can no longer be evaluated in the closed form (32)–(33), they are represented by their own approximators, as are the cross-channel maps (47):

$$\hat{u}_i = \hat{W}_{u_i}^T \psi_{u_i}(\delta_i), \quad \hat{\vartheta}_i = \hat{W}_{\vartheta_i}^T \psi_{\vartheta_i}(\delta_i), \quad \Xi_{ij}^u = \hat{W}_{ij}^{uT} \psi_{ij}(\delta), \quad \Xi_{ij}^\vartheta = \hat{W}_{ij}^{\vartheta T} \psi_{ij}(\delta). \tag{49}$$

Every unknown in (48) then enters *linearly*, so the entire estimation problem retains the structure of Section 4.2. Collecting

$$\mathcal{Z}_i \triangleq \big[\hat{W}_i^T, \ \mathrm{vec}(\hat{W}_{u_i})^T, \ \mathrm{vec}(\hat{W}_{\vartheta_i})^T, \ \{\mathrm{vec}(\hat{W}_{ij}^u)^T\}_{j \in \mathcal{N}_i}, \ \{\mathrm{vec}(\hat{W}_{ij}^\vartheta)^T\}_{j \in \mathcal{N}_i}\big]^T \tag{50}$$

and, using $a^T X b = (b^T \otimes a^T)\mathrm{vec}(X)$, the matching regressor

$$\Lambda_i(t) \triangleq \Big[\Delta\phi_i(t)^T,\ 2\int_{t-T}^{t}\big((u_i-\hat{u}_i)^T R_{ii}\otimes\psi_{u_i}^T\big)d\tau,\ -2\gamma^2\int_{t-T}^{t}\big((\vartheta_i-\hat{\vartheta}_i)^T T_{ii}\otimes\psi_{\vartheta_i}^T\big)d\tau,$$

$$\Big\{\int_{t-T}^{t}\big((u_j-\hat{u}_j)^T\otimes\psi_{ij}^T\big)d\tau\Big\}_{j\in\mathcal{N}_i},\ \Big\{\int_{t-T}^{t}\big((\vartheta_j-\hat{\vartheta}_j)^T\otimes\psi_{ij}^T\big)d\tau\Big\}_{j\in\mathcal{N}_i}\Big]^T, \quad (51)$$

equation (48) becomes $\Lambda_i(t)^T\mathcal{Z}_i+\varrho_i(t)=\epsilon_{B_i}(t)$ along the realized trajectory, with ϱ_i unchanged. Consequently the model-free Bellman residual is

$$\Psi_i^{\mathrm{mf}}(t) \triangleq \Lambda_i(t)^T\mathcal{Z}_i+\varrho_i(t), \quad (52)$$

and the adaptive law (35), the experience-replay mechanism, and the error dynamics (43) all carry over *verbatim* under the substitution

$$\Delta\phi_i \longmapsto \Lambda_i, \qquad \hat{W}_i \longmapsto \mathcal{Z}_i, \qquad \Psi_i \longmapsto \Psi_i^{\mathrm{mf}}, \qquad \Theta_i \longmapsto \Theta_i^{\mathrm{mf}}, \quad (53)$$

with Θ_i^{mf} given by (55) below. In particular Ξ_i, $\mathcal{E}_i$ and $\mathcal{T}_i$ of (40)–(42) keep their definitions with $\Delta\bar{\phi}_i$ built from Λ_i instead of $\Delta\phi_i$. *The model-free extension is thus a change of regressor, not a change of algorithm* – which is why the mean-square UUB result of Theorem 1 applies to it unchanged.

Model-free stabilizing term. The term Θ_i of (37) is the gradient, with respect to the weights being adapted, of the control-channel contribution to $\delta_i^T\dot{\delta}_i$; in the parameterisation (49) that gradient is taken with respect to $\mathrm{vec}(\hat{W}_{u_i})$ and $\mathrm{vec}(\hat{W}_{\vartheta_i})$ rather than $\hat{W}_i$, and therefore requires the input channel only through the *vector-valued* quantities

$$\Upsilon_i \triangleq (b_i+d_i)g_i^T(x_i)\delta_i \in \mathbb{R}^{m_i}, \qquad \Upsilon_i^{\vartheta} \triangleq (b_i+d_i)k_i^T(x_i)\delta_i \in \mathbb{R}^{q_i}, \quad (54)$$

together with their neighbour counterparts $\Upsilon_{ij} \triangleq a_{ij}g_j^T\delta_i$, $\Upsilon_{ij}^{\vartheta} \triangleq a_{ij}k_j^T\delta_i$. These are identified from the *same* off-policy data set by applying the behaviour–target split (44) to $S_i(\delta_i) \triangleq \frac{1}{2}\delta_i^T\delta_i$ instead of to V_i: since $\nabla S_i=\delta_i$ is measured, Itô's formula gives $\delta_i^T\Delta_i=\Upsilon_i^T(u_i-\hat{u}_i)+\Upsilon_i^{\vartheta T}(\vartheta_i-\hat{\vartheta}_i)-\sum_{j\in\mathcal{N}_i}\big(\Upsilon_{ij}^T(u_j-\hat{u}_j)+\Upsilon_{ij}^{\vartheta T}(\vartheta_j-\hat{\vartheta}_j)\big)$, a scalar identity linear in the unknowns of (54) and driven by the same measured mismatch signals as (48). Note that this auxiliary problem is small – Υ_i has the dimension of the input, not of the critic basis – precisely because ∇S_i is known whereas $\nabla\phi_i$ is not. The model-free stabilizing term is then

$$\Theta_i^{\mathrm{mf}}(t) = -\beta_i\int_{t-T}^{t}\Big[\mathbf{0}_L;\ \tfrac{1}{2}\Upsilon_i\otimes\psi_{u_i};\ -\tfrac{1}{2\gamma^2}\Upsilon_i^{\vartheta}\otimes\psi_{\vartheta_i};\ \big\{\tfrac{1}{2}\Upsilon_{ij}\otimes\psi_{ij}\big\}_j;\ \big\{-\tfrac{1}{2\gamma^2}\Upsilon_{ij}^{\vartheta}\otimes\psi_{ij}\big\}_j\Big]d\tau, \quad (55)$$

which is block-conformable with $\mathcal{Z}_i$ in (50) and reduces to (37) whenever the actor bases are chosen as $\psi_{u_i}=R_{ii}^{-1}g_i^T\nabla\phi_i^T$, i.e. when the model is in fact known. Its leading block is zero because, under (49), the critic weights $\hat{W}_i$ no longer enter the control channel directly – the actors do. Since $\Upsilon_i,\Upsilon_i^{\vartheta}$ are bounded on the compact set Ω whenever δ_i is and the activation functions are bounded by Assumption 4, Θ_i^{mf} satisfies the same bound $\|\Theta_i^{\mathrm{mf}}\| \le \theta_{max_i}$ used throughout the proof of Theorem 1.

Excitation requirement. Condition 1 must be restated for the enlarged regressor: the history stack $Z_i=\{\Lambda_i(t_1),\ldots,\Lambda_i(t_l)\}$ is required to satisfy $\mathrm{rank}(Z_i)=\dim(\mathcal{Z}_i)=L+m_iL_{u_i}+q_iL_{\vartheta_i}+\sum_{j\in\mathcal{N}_i}(m_j+q_j)L_{ij}$, with $l>\dim(\mathcal{Z}_i)$, which is the concurrent-learning analogue of the rank condition on $[I_{xx},I_{xu},I_{xv},\delta_{uu}]$ imposed in [21, Lemma 5]. As in the model-based case this is a verifiable condition on recorded data rather than an a priori PE requirement on the trajectory; the price of removing the model is that the exploration signals e_j must be rich enough to excite the enlarged regressor, so more data are required than in Condition 1.

The resulting procedure is summarized in Algorithm 2. Comparing it with Algorithm 1 makes the nature of the extension explicit: the two differ only in how the regressor is built (steps 3–8), while the learning mechanism – residual, history stack, concurrent-learning update – is untouched.

Algorithm 2 Model-Free Stochastic IRL for the H_∞ Graphical Game

Require: $Q_{ii} \succeq 0$, $R_{ij}, T_{ij} \succ 0$, $\gamma > 0$, $\alpha_i, \beta_i > 0$, $T > 0$; bases $\phi_i, \psi_{u_i}, \psi_{\vartheta_i}, \{\psi_{ij}\}_{j\in\mathcal{N}_i}$; admissible $\mathcal{Z}_i(0)$; stack size $l > \dim(\mathcal{Z}_i)$.
Data collection (off-policy). Apply $u_j = \hat{u}_j + e_j$, $j \in \{i\} \cup \mathcal{N}_i$, with e_j bounded and satisfying the rank condition of Section 4.3.

loop for each t
  Measure $\delta_i(t), u_i(t), \vartheta_i(t)$; receive $\{\delta_j, u_j, \vartheta_j, \hat{u}_j, \hat{\vartheta}_j\}_{j\in\mathcal{N}_i}$.
  $\hat{u}_i \leftarrow \hat{W}_{u_i}^T \psi_{u_i}(\delta_i), \quad \hat{\vartheta}_i \leftarrow \hat{W}_{\vartheta_i}^T \psi_{\vartheta_i}(\delta_i)$ ▷ cf. (49); no g_i, k_i
  $\Delta\phi_i(t) \leftarrow \phi_i(\delta_i(t)) - \phi_i(\delta_i(t-T)), \quad \varrho_i(t) \leftarrow \int_{t-T}^{t} r_i \, d\tau$
  $\Lambda_i(t) \leftarrow$ (51), assembled from $\Delta\phi_i(t)$ and the mismatches $u_j - \hat{u}_j,\ \vartheta_j - \hat{\vartheta}_j$
  $\Psi_i^{\mathrm{mf}}(t) \leftarrow \Lambda_i(t)^T \mathcal{Z}_i + \varrho_i(t)$
  $\xi_i(t) \leftarrow \mathbb{1}\{\delta_i^T(t)\delta_i(t) > \delta_i^T(t-T)\delta_i(t-T)\}$
  $\{\Upsilon_i, \Upsilon_i^\vartheta, \Upsilon_{ij}, \Upsilon_{ij}^\vartheta\} \leftarrow$ least squares on the $\frac{1}{2}\delta_i^T\delta_i$ identity; $\Theta_i^{\mathrm{mf}}(t) \leftarrow$ (55)
  $Z_i \leftarrow Z_i \cup \{\Lambda_i(t)\}$ while $\operatorname{rank}(Z_i) < \dim(\mathcal{Z}_i)$

$$\dot{\mathcal{Z}}_i \leftarrow -\alpha_i \frac{\Lambda_i(t)\Psi_i^{\mathrm{mf}}(t)}{\left(\Lambda_i^T(t)\Lambda_i(t)+1\right)^2} + \xi_i(t)\Theta_i^{\mathrm{mf}}(t) + \sum_{k=1}^{l}\left(-\alpha_i \frac{\Lambda_i(t_k)\Psi_i^{\mathrm{mf}}(t_k)}{\left(\Lambda_i^T(t_k)\Lambda_i(t_k)+1\right)^2} + \xi_i(t_k)\Theta_i^{\mathrm{mf}}(t_k)\right)$$

end loop

Ensure: $\mathcal{Z}_i \to \mathcal{Z}_i^*$, $\hat{V}_i = \hat{W}_i^T\phi_i$, $\hat{u}_i, \hat{\vartheta}_i$ mean-square UUB (Corollary 1), with $f_{ei}, g_i, k_i, \Sigma_i$ all unknown.

Remark 3. *Unlike [21], where the control-dependent diffusion $Cx + Du$ forces the extra lumped unknown D^TPD into the least-squares problem, the process-noise model (5) makes Σ_i independent of u_i and ϑ_i. The Itô correction $\frac{1}{2}\mathrm{tr}(\Sigma_i^T\nabla^2 V_i\Sigma_i)$ is therefore common to $\hat{F}_i$ and cancels with the Bellman identity, so* no *diffusion-dependent parameter has to be identified: Algorithm 2 is model-free in the diffusion channel for free. The same structural reason is why the mixed $\mathcal{H}_2/\mathcal{H}_\infty$ scheme of Cui and Molu must assume the noise feed-through matrix known, whereas the present scheme need not.*

4.4 Stability and convergence analysis

One constant used below must be introduced explicitly, since it does not follow from Assumption 4.

Assumption 5. *Let $F_i \triangleq f_{ei} + (b_i + d_i)(g_i\hat{u}_i + k_i\hat{\vartheta}_i) - \sum_{j\in\mathcal{N}_i} a_{ij}(g_j\hat{u}_j + k_j\hat{\vartheta}_j)$ denote the closed-loop drift of (11) under the approximate policies actually applied. On the compact set Ω there exists a constant $\zeta_i > 0$ such that, whenever $\delta_i^T(t)\delta_i(t) \le \delta_i^T(t-T)\delta_i(t-T)$ (*strict decrease on the contracting regime*),*

$$\mathbb{E}\Big[\int_{t-T}^{t} \delta_i^T F_i \, d\tau \,\Big|\, \xi_i(t) = 0\Big] \le -\zeta_i \, \mathbb{E}\Big[\int_{t-T}^{t} \|\delta_i(\tau)\| \, d\tau \,\Big|\, \xi_i(t) = 0\Big]. \tag{56}$$

Remark 4. *This is a genuine assumption, not a consequence of $\xi_i = 0$: as (63) makes explicit, the switching rule alone – without further hypotheses – yields only that the drift term and an uncontrolled, conditioned martingale term sum to something non-positive, not a bound on the drift term in isolation. Assumption 5 instead posits directly, as a hypothesis on the closed-loop dynamics, that a strict decrease proportional to $\mathbb{E}\int\|\delta_i\|$ holds on this event; this is the continuous-time analogue of the detectability-type conditions imposed on the policy-evaluation step in [21], and holds, for instance, whenever $\hat{u}_i$ is admissible in the sense of Definition 1 with a uniform decay rate on $\Omega \setminus \{0\}$. Unlike an earlier draft of this result, no separate "closed-loop linear growth" bound on $\delta_i^T F_i$ is required: the $\xi_i(t) = 1$ regime below bounds the corresponding window term directly from the elementary non-negativity of a squared norm, without any reference to F_i.*

Theorem 1. *Given the stochastic system dynamics in (11) and the coupled stochastic HJI equations in (22), suppose the control law and disturbance policy are generated by (32) and (33), respectively. Assume that the update law for the critic NN weights W_i follows (35), using the realized single-trajectory residual $\epsilon_{B_i}(t)$ of (29). If **Condition** 1 and Assumption 5 hold, then the closed-loop tracking error δ_i and the weight estimation error $\tilde{W}_i$ are ensured to be ultimately uniformly bounded* in the mean-square sense *(mean-square UUB): $\mathbb{E}\|\delta_i(t)\|^2$ and $\mathbb{E}\|\tilde{W}_i(t)\|^2$ are ultimately bounded, with an ultimate bound that reduces exactly to the pathwise UUB bound of the deterministic case as the diffusion intensity $\Sigma_i \to 0$. Furthermore, the approximated value function $\hat{V}_i$, the control policy $\hat{u}_i$, and the worst-case disturbance approach their corresponding optimal solutions in mean square: $\mathbb{E}\|V_i^* - \hat{V}_i\|^2 < \epsilon_{V_i}$, $\mathbb{E}\|u_i^* - \hat{u}_i\|^2 < \epsilon_{u_i}$, and $\mathbb{E}\|\vartheta_i^* - \hat{\vartheta}_i\|^2 < \epsilon_{\vartheta_i}$, where ϵ_{V_i}, ϵ_{u_i}, and ϵ_{ϑ_i} are the small positive constants (76)–(77). These constants decrease to their deterministic values as $\Sigma_i \to 0$ and to zero only as the critic basis is refined, $L \to \infty$; they do not vanish with the noise alone, since the neural-network approximation error persists in the noise-free limit.*

Proof. Two departures from the deterministic proof must be made at the outset, both caused by the switching signal ξ_i becoming a *random* variable once δ_i is an Itô process.

First, the deterministic argument uses one Lyapunov function on $\{\xi_i = 0\}$ and a different one on $\{\xi_i = 1\}$. Switching between two Lyapunov functions is legitimate only under a multiple-Lyapunov-function (dwell-time) argument that accounts for the jump $L_i^{(1)} - L_i^{(0)}$ at each switching instant. We avoid that difficulty altogether by using the *single* function

$$L_i(t) \triangleq \frac{1}{2}\int_{t-T}^{t} \delta_i^T \delta_i \, d\tau + \frac{1}{2}\tilde{W}_i^T \tilde{W}_i + V_i(\delta_i(t)) \tag{57}$$

in *both* regimes, where $V_i \geq 0$ is a local smooth solution of the coupled stochastic HJI (22). Since $V_i \geq 0$, (57) dominates the windowed function used in the deterministic $\xi_i = 0$ analysis, so no conclusion is weakened; the two regimes now differ only in how $\mathbb{E}[\dot{L}_i]$ is *bounded*, not in which function is being bounded.

Second, the events $\{\xi_i(t) = 0\}$ and $\{\xi_i(t) = 1\}$ are $\mathcal{F}_t$-measurable events of positive probability rather than a partition of the time axis: at a fixed t, some sample paths contract over $[t-T, t]$ and others do not. Taking an *unconditional* expectation of an inequality that holds only on one of these events is therefore invalid. All expectations in the two regimes below are accordingly conditional, and we abbreviate $\mathbb{E}_0[\,\cdot\,] \triangleq \mathbb{E}[\,\cdot \,|\, \xi_i(t) = 0]$ and $\mathbb{E}_1[\,\cdot\,] \triangleq \mathbb{E}[\,\cdot \,|\, \xi_i(t) = 1]$. They are recombined at the end by the tower property

$$\mathbb{E}[\dot{L}_i(t)] = \mathbb{P}\big(\xi_i(t) = 0\big)\, \mathbb{E}\big[\dot{L}_i(t) \,\big|\, \xi_i(t) = 0\big] + \mathbb{P}\big(\xi_i(t) = 1\big)\, \mathbb{E}\big[\dot{L}_i(t) \,\big|\, \xi_i(t) = 1\big], \tag{58}$$

so that a bound established in each regime yields a bound on $\mathbb{E}[\dot{L}_i(t)]$ as a convex combination, with no additional hypothesis on the switching statistics.

With (57) fixed, two further facts must be kept separate. First, the window integral $\frac{1}{2}\int_{t-T}^{t} \delta_i^T \delta_i \, d\tau$ is differentiated in t by the ordinary Leibniz rule for a moving window, which requires only a.s. continuity of the (now stochastic) integrand $\delta_i^T(\tau)\delta_i(\tau)$ – true since (11) has continuous sample paths – and gives, pathwise, $\frac{d}{dt}\big[\frac{1}{2}\int_{t-T}^{t} \delta_i^T \delta_i \, d\tau\big] = \frac{1}{2}\big[\delta_i^T(t)\delta_i(t) - \delta_i^T(t-T)\delta_i(t-T)\big]$, exactly as in the deterministic case. Second, unlike the deterministic case, this boundary term can no longer be rewritten as $\int_{t-T}^{t} \delta_i^T \dot{\delta}_i \, d\tau$ by the ordinary fundamental theorem of calculus, since $\delta_i(\tau)$ has no pathwise derivative; Itô's formula must be used instead. Applying Itô's formula to $\delta_i^T(\tau)\delta_i(\tau)$ along (11) gives

$$\tfrac{1}{2}\big[\delta_i^T(t)\delta_i(t) - \delta_i^T(t-T)\delta_i(t-T)\big] = \int_{t-T}^{t} \delta_i^T F_i \, d\tau + \tfrac{1}{2}\int_{t-T}^{t} \operatorname{tr}\big(\Sigma_i^T \Sigma_i\big)\, d\tau + \int_{t-T}^{t} \delta_i^T \Sigma_i \, d\mathcal{W}_i(\tau), \tag{59}$$

where F_i is the closed-loop drift defined in Assumption 5, the middle term is the new Itô correction, and the last term is a local martingale that is conditionally mean-zero and vanishes upon taking expectations. The third summand of (57) is handled by Itô's formula applied to $V_i(\delta_i(t))$ along (11), giving $dV_i = \mathcal{A}_i^{\hat{u}} V_i\, d\tau + \nabla V_i^T \Sigma_i\, d\mathcal{W}_i$, in which $\mathcal{A}_i^{\hat{u}}$ is the generator (18) evaluated at the *approximate* policies actually applied. The HJI equation (19) certifies $\mathcal{A}_i^{u^*} V_i = -r_i$ at the *optimal* policies, so the two differ by a policy-mismatch term,

$$\mathcal{A}_i^{\hat{u}} V_i = -r_i + \nabla V_i^T \Big[(b_i + d_i)\big(g_i(\hat{u}_i - u_i^*) + k_i(\hat{\vartheta}_i - \vartheta_i^*)\big) - \sum_{j\in\mathcal{N}_i} a_{ij}\big(g_j(\hat{u}_j - u_j^*) + k_j(\hat{\vartheta}_j - \vartheta_j^*)\big)\Big], \quad (60)$$

which is precisely the source of the Γ- and Λ-dependent terms collected below; the Itô trace term is common to both generators and cancels from the difference. Combining with $\mathbb{E}[\tilde{W}_i^T \dot{\tilde{W}}_i]$ (unaffected, since $\hat{W}_i$ solves the ordinary differential recursion (35) driven by the realized trajectory), the exact mean rate of (57) is

$$\mathbb{E}[\dot{L}_i(t)] = \mathbb{E}\Big[\int_{t-T}^{t} \delta_i^T F_i\, d\tau + \tilde{W}_i^T \dot{\tilde{W}}_i + \mathcal{A}_i^{\hat{u}} V_i\Big] + \frac{1}{2}\mathbb{E}\Big[\int_{t-T}^{t} \mathrm{tr}\big(\Sigma_i^T \Sigma_i\big)\, d\tau\Big], \quad (61)$$

which we bound above by

$$\mathbb{E}[\dot{L}_i(t)] \le \mathbb{E}\Big[\int_{t-T}^{t} \delta_i^T F_i\, d\tau + \tilde{W}_i^T \dot{\tilde{W}}_i + \mathcal{A}_i^{\hat{u}} V_i\Big] + c_{\sigma_i} T, \quad (62)$$

where $c_{\sigma_i} \triangleq \frac{1}{2}\sup_{x\in\Omega} \mathrm{tr}(\Sigma_i(x)^T \Sigma_i(x))$ is bounded by Assumption 4, so that $c_{\sigma_i} T \ge \frac{1}{2}\mathbb{E}\big[\int_{t-T}^{t} \mathrm{tr}(\Sigma_i^T \Sigma_i)\, d\tau\big]$ for every window of length T. Note $c_{\sigma_i} \to 0$ exactly as $\Sigma_i \to 0$, recovering the deterministic $\dot{L}_i(t)$ in that limit. Every expectation from here until (58) is conditional on the regime under discussion.

Regime $\xi_i(t) = 0$. On this event $\delta_i^T(t)\delta_i(t) \le \delta_i^T(t-T)\delta_i(t-T)$, so the left-hand side of (59) is non-positive. Taking $\mathbb{E}[\,\cdot\,|\,\xi_i(t) = 0]$ of (59) and dropping the non-negative trace term gives

$$\mathbb{E}_0\Big[\int_{t-T}^{t} \delta_i^T F_i\, d\tau\Big] + \mathbb{E}_0\Big[\int_{t-T}^{t} \delta_i^T \Sigma_i\, d\mathcal{W}_i(\tau)\Big] \le 0. \quad (63)$$

Unlike the deterministic case, the martingale term here need *not* vanish upon conditioning: $\{\xi_i(t) = 0\}$ is an $\mathcal{F}_t$-measurable event that depends on $\delta_i(t)$, hence on the very martingale increment being conditioned on, so the usual "martingale $\Rightarrow$ zero conditional mean" property – valid for conditioning on an $\mathcal{F}_{t-T}$-measurable event – does not apply here. Consequently (63) only certifies that the drift term and the (generally nonzero, sign-uncontrolled) conditioned martingale term *together* sum to something non-positive, not that the drift term does so on its own; it is retained here only as motivation for the switching rule, not as a derivation step. The strict, quantified decrease actually used below is instead supplied directly by Assumption 5, i.e. by (56), which is posited exactly in the conditional form needed and does not rely on (63).

Two bounds are needed before substituting. First, from (60) and the definition of r_i, the generator along the applied policies obeys

$$\mathcal{A}_i^{\hat{u}} V_i \;\le\; -Q_{ii}(\delta_i) + \varepsilon_{\pi_i} \;\le\; -\lambda_{min}(Q_{ii})\|\delta_i\|^2 + \varepsilon_{\pi_i}, \qquad \varepsilon_{\pi_i} \triangleq \sup_{\Omega}\Big[\mathcal{A}_i^{\hat{u}} V_i + Q_{ii}(\delta_i)\Big], \quad (64)$$

which is finite on the compact set Ω by Assumption 4. The constant ε_{π_i} absorbs both the policy-mismatch bracket of (60) *and* the sign-indefinite disturbance-cost terms $+\gamma^2\vartheta_i^T T_{ii}\vartheta_i + \gamma^2\sum_j \vartheta_j^T T_{ij}\vartheta_j$ contained in $-r_i$; it vanishes when $\hat{u}_i \to u_i^*$, $\hat{\vartheta}_i \to \vartheta_i^*$ and $\vartheta_i \to 0$. Second, $-\alpha_i\mathbb{E}_0[\tilde{W}_i^T \Xi_i \tilde{W}_i] \le -\alpha_i\lambda_{min}(\Xi_i)\mathbb{E}_0\|\tilde{W}_i\|^2$, with $\lambda_{min}(\Xi_i) > 0$ by Condition 1. Substituting these together with (43) and (56),

$$\mathbb{E}_0\big[\dot{L}_i(t)\big] \le -\zeta_i\,\mathbb{E}_0\Big[\int_{t-T}^{t} \|\delta_i(\tau)\|d\tau\Big] - \lambda_{min}(Q_{ii})\,\mathbb{E}_0\|\delta_i\|^2 - \alpha_i\lambda_{min}(\Xi_i)\,\mathbb{E}_0\|\tilde{W}_i\|^2 + \big[\alpha_i(l+1)\epsilon_{max} + l\,\theta_{max_i}\big]\,\mathbb{E}_0\|\tilde{W}_i\| + \varepsilon_{\pi_i} + c_{\sigma_i}T, \tag{65}$$

where, since $\xi_i(t)=0$ on this event, $\mathcal{T}_i(t)=\sum_{k=1}^{l}\xi_i(t_k)\Theta_i(t_k)$ contains only the historical sum, so the Cauchy–Schwarz bound after (43) sharpens to $-\mathbb{E}_0[\tilde{W}_i^T\mathcal{T}_i(t)] \le l\,\theta_{max_i}\,\mathbb{E}_0\|\tilde{W}_i\|$ (one fewer term than the general $(l+1)\theta_{max_i}$ bound), while $\alpha_i\,\mathbb{E}_0[\tilde{W}_i^T\mathcal{E}_i(t)] \le \alpha_i(l+1)\epsilon_{max}\,\mathbb{E}_0\|\tilde{W}_i\|$ as before. The first three terms of (65) are non-positive and act on $\mathbb{E}_0\int\|\delta_i\|$, $\mathbb{E}_0\|\delta_i\|^2$ and $\mathbb{E}_0\|\tilde{W}_i\|^2$ respectively, while the remaining terms are the perturbation. Hence $\mathbb{E}_0[\dot{L}_i(t)]<0$ holds as soon as *any one* of the three dominates that perturbation, in particular whenever

$$\mathbb{E}_0\Big[\int_{t-T}^{t} \|\delta_i(\tau)\|d\tau\Big] \ge \frac{\big[\alpha_i(l+1)\epsilon_{max} + l\,\theta_{max_i}\big]\mathbb{E}_0\|\tilde{W}_i\| + \varepsilon_{\pi_i} + c_{\sigma_i}T}{\zeta_i} \triangleq \hat{\chi}_{1i} \tag{66}$$

or, completing the square in $\mathbb{E}_0\|\tilde{W}_i\|$ in the third and fourth terms,

$$\mathbb{E}_0\|\tilde{W}_i\| \ge \frac{\alpha_i(l+1)\epsilon_{max} + l\,\theta_{max_i}}{\alpha_i\lambda_{\min}(\Xi_i)} + \sqrt{\frac{\varepsilon_{\pi_i} + c_{\sigma_i}T}{\alpha_i\lambda_{\min}(\Xi_i)}} \triangleq \chi_{1i} \tag{67}$$

where $\lambda_{\min}(\Xi_i)$ is the smallest eigenvalue of Ξ_i. Both thresholds reduce to their deterministic counterparts as $c_{\sigma_i}\to 0$; the θ_{max_i}, ϵ_{max} and ε_{π_i} contributions are approximation-level terms that persist in the deterministic limit.

Regime $\xi_i(t)=1$. Here the boundary term of (59) is positive, so the contraction (63) is unavailable and the δ_i-negativity must instead come from the V_i summand of the *same* Lyapunov function (57) – this is where carrying $V_i\ge 0$ in both regimes pays off, since no change of Lyapunov function, and hence no jump term, is incurred at a switching instant.

The window term is bounded without reference to F_i, Assumption 5, or any window/T-dependent quantity at all: since $\delta_i^T(t-T)\delta_i(t-T)\ge 0$ always, the exact (Leibniz) identity for this term gives, pathwise,

$$\frac{d}{dt}\Big[\frac{1}{2}\int_{t-T}^{t} \delta_i^T\delta_i\,d\tau\Big] = \frac{1}{2}\delta_i^T(t)\delta_i(t) - \frac{1}{2}\delta_i^T(t-T)\delta_i(t-T) \;\le\; \frac{1}{2}\|\delta_i(t)\|^2. \tag{68}$$

For the V_i summand, expanding $\mathcal{A}_i^{\hat{u}}V_i$ by (60) and writing $r_i \triangleq Q_{ii}(\delta_i) + u_i^TR_{ii}u_i + \sum_{j\in\mathcal{N}_i}u_j^TR_{ij}u_j - \gamma^2\vartheta_i^TT_{ii}\vartheta_i - \gamma^2\sum_{j\in\mathcal{N}_i}\vartheta_j^TT_{ij}\vartheta_j$ for the instantaneous cost of (13), and $\Gamma_i \triangleq g_iR_{ii}^{-1}g_i^T$, $\Gamma_j \triangleq \sum_{j\in\mathcal{N}_i}g_jR_{jj}^{-1}R_{ij}R_{jj}^{-1}g_j^T$, $\Lambda_i \triangleq k_iT_{ii}^{-1}k_i^T$, $\Lambda_j \triangleq \sum_{j\in\mathcal{N}_i}k_jT_{jj}^{-1}T_{ij}T_{jj}^{-1}k_j^T$ with $\Gamma_{i_{min}} \le \|\Gamma_i\| \le \Gamma_{i_{max}}$, $\Gamma_{j_{min}} \le \|\Gamma_j\| \le \Gamma_{j_{max}}$, $\Lambda_{i_{min}} \le \|\Lambda_i\| \le \Lambda_{i_{max}}$, $\Lambda_{j_{min}} \le \|\Lambda_j\| \le \Lambda_{j_{max}}$, using $-Q_{ii}(\delta_i) \le -\lambda_{min}(Q_{ii})\mathbb{E}\|\delta_i\|^2$ and the worst-case bounds on $u_i^TR_{ii}u_i + \sum_j u_j^TR_{ij}u_j$ and $\vartheta_i^TT_{ii}\vartheta_i + \sum_j\vartheta_j^TT_{ij}\vartheta_j$ implied by the feedback forms (32)–(33), the generator obeys

$$\mathcal{A}_i^{\hat{u}}V_i \le -\lambda_{min}(Q_{ii})\|\delta_i\|^2 + \tfrac{1}{2}b_{\nabla\phi_i}\big(\gamma^2\Lambda_{i_{max}} - \Gamma_{i_{min}}\big)\|\delta_i\| + \tfrac{1}{2}b_{\nabla\phi_j}\big(\gamma^2\Lambda_{j_{max}} - \Gamma_{j_{min}}\big)\|\delta_j\| + c_{V_i}, \tag{69}$$

where $c_{V_i} \triangleq \frac{1}{2}\sup_{x\in\Omega}\operatorname{tr}\big(\Sigma_i(x)^T\nabla^2V_i(\delta_i)\,\Sigma_i(x)\big)$ is the Itô-correction term of the generator (18) itself, evaluated at the applied policies – finite on Ω by Assumption 4 (which bounds $\nabla^2\phi_i, \nabla^2\epsilon_i, \Sigma_i$, and hence $\nabla^2V_i = W_i^T\nabla^2\phi_i + \nabla^2\epsilon_i$) – and *distinct* from c_{σ_i} of (62): c_{σ_i} bounds the unrelated Hessian-of-$\delta_i^T\delta_i$ trace term $\operatorname{tr}(\Sigma_i^T\Sigma_i)$ used only in the windowed identity (59) of Regime $\xi_i(t)=0$, whereas c_{V_i} bounds the trace term intrinsic to $\mathcal{A}_i^{\hat{u}}V_i$ via ∇^2V_i. Like c_{σ_i}, $c_{V_i}\to 0$ exactly as $\Sigma_i\to 0$; and since (69) is the exact pointwise generator evaluated at t (not an integral over $[t-T,t]$), no window/T-dependent bound is needed here, unlike in Regime $\xi_i(t) = 0$. Combining (68) and (69) with (43),

$$\begin{aligned}\mathbb{E}_1\big[\dot{L}_i(t)\big] \le \Big(\tfrac{1}{2} - \lambda_{min}(Q_{ii})\Big)\mathbb{E}_1\|\delta_i\|^2 + \tfrac{1}{2}b_{\nabla\phi_i}\big(\gamma^2\Lambda_{i_{max}} - \Gamma_{i_{min}}\big)\mathbb{E}_1\|\delta_i\| \\ + \tfrac{1}{2}b_{\nabla\phi_j}\big(\gamma^2\Lambda_{j_{max}} - \Gamma_{j_{min}}\big)\mathbb{E}_1\|\delta_j\| - \alpha_i\mathbb{E}_1\big[\tilde{W}_i^T\Xi_i\tilde{W}_i\big] \\ + \alpha_i\,\mathbb{E}_1\big[\tilde{W}_i^T\mathcal{E}_i(t)\big] - \mathbb{E}_1\big[\tilde{W}_i^T\mathcal{T}_i(t)\big] + c_{V_i}\end{aligned} \tag{70}$$

Applying the identity $x = (x+\frac{1}{2})^2 - x^2 - \frac{1}{4}$ to the $\mathbb{E}\|\delta_i\|$-linear term, so that it combines with $\mathbb{E}\|\delta_i\|^2$ into a single square, and bounding the $\tilde{W}_i$-cross terms by the Cauchy–Schwarz inequalities established after (43), gives the fully collected bound

$$\begin{aligned}\mathbb{E}_1\big[\dot{L}_i(t)\big] \le \Big(\tfrac{1}{2} - \lambda_{min}(Q_{ii}) + \tfrac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{min}} - \tfrac{1}{2}\gamma^2 b_{\nabla\phi_i}\Lambda_{i_{max}}\Big)\mathbb{E}_1\|\delta_i\|^2 \\ + \Big(\tfrac{1}{2}\gamma^2 b_{\nabla\phi_i}\Lambda_{i_{max}} - \tfrac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{min}}\Big)\Big(\mathbb{E}_1\|\delta_i\| + \tfrac{1}{2}\Big)^2 + \tfrac{1}{8}b_{\nabla\phi_i}\big(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}}\big) \\ + \tfrac{1}{2}b_{\nabla\phi_j}\big(\gamma^2\Lambda_{j_{max}} - \Gamma_{j_{min}}\big)\mathbb{E}_1\|\delta_j\| - \alpha_i\lambda_{min}(\Xi_i)\,\mathbb{E}_1\|\tilde{W}_i\|^2 \\ + \big[\alpha_i(l+1)\epsilon_{max} + (l+1)\theta_{max_i}\big]\,\mathbb{E}_1\|\tilde{W}_i\| + c_{V_i}\end{aligned} \tag{71}$$

To ensure the negativity of $\mathbb{E}_1[\dot{L}_i(t)]$, writing $D_i \triangleq 2\lambda_{\min}(Q_{ii}) - 1 + \gamma^2 b_{\nabla\phi_i}\Lambda_{i_{max}} - b_{\nabla\phi_i}\Gamma_{i_{min}} > 0$ for twice the magnitude of the (negative) $\mathbb{E}\|\delta_i\|^2$ coefficient in (71), it is required that

$$\mathbb{E}_1\|\delta_i\|^2 \ge \frac{b_{\nabla\phi_i}\big(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}}\big)}{4\,D_i} + \frac{2\,c_{V_i}}{D_i} \triangleq \hat{\chi}_{2i} \tag{72}$$

or

$$\mathbb{E}_1\|\tilde{W}_i\| \ge \frac{\alpha_i(l+1)\epsilon_{max} + (l+1)\theta_{max_i}}{\alpha_i\lambda_{min}(\Xi_i)} + \sqrt{\frac{c_{V_i} + \frac{1}{8}b_{\nabla\phi_i}\big(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}}\big)}{\alpha_i\lambda_{min}(\Xi_i)}} \triangleq \chi_{2i} \tag{73}$$

The first term of (72) is the constant $\frac{1}{8}b_{\nabla\phi_i}(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}})$ of (71) divided by $\frac{1}{2}D_i$, and the second is c_{V_i} divided by the same quantity; the numerator carries the full factor $(\Gamma_{i_{min}} - \gamma^2\Lambda_{i_{max}})$, positive under the sign conditions imposed next. Those conditions, which also let the sign-indefinite $\big(\mathbb{E}_1\|\delta_i\| + \frac{1}{2}\big)^2$ term and the linear $\mathbb{E}_1\|\delta_j\|$ term of (71) be dropped as non-positive, are

$$\left(\frac{1}{2} - \lambda_{min}(Q_{ii}) + \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{min}} - \frac{1}{2}\gamma^2 b_{\nabla\phi_i}\Lambda_{i_{max}}\right) < 0,$$

$$\left(\frac{1}{2}\gamma^2 b_{\nabla\phi_i}\Lambda_{i_{max}} - \frac{1}{2}b_{\nabla\phi_i}\Gamma_{i_{min}}\right) < 0, \qquad \left(\frac{1}{2}\gamma^2 b_{\nabla\phi_j}\Lambda_{j_{max}} - \frac{1}{2}b_{\nabla\phi_j}\Gamma_{j_{min}}\right) < 0$$

Recombination. Both regimes bound the mean rate of the *same* function (57), so by the tower property (58) the unconditional rate $\mathbb{E}[\dot{L}_i(t)]$ is a convex combination of the two conditional rates and is therefore negative whenever both are. Collecting (66)–(67) and (72)–(73), $\mathbb{E}[\dot{L}_i(t)] < 0$ holds outside the set on which

$$\mathbb{E}_1\|\delta_i\|^2 < \hat{\chi}_{2i}, \quad \mathbb{E}_0\Big[\int_{t-T}^{t} \|\delta_i(\tau)\|\,d\tau\Big] < \hat{\chi}_{1i}, \quad \mathbb{E}_0\|\tilde{W}_i\| < \chi_{1i}, \quad \mathbb{E}_1\|\tilde{W}_i\| < \chi_{2i} \tag{74}$$

all hold simultaneously. Since $\mathbb{E}\|\tilde{W}_i\| \le \max\{\mathbb{E}_0\|\tilde{W}_i\|, \mathbb{E}_1\|\tilde{W}_i\|\}$ by (58), the weight-error bound aggregates to the unconditional statement $\mathbb{E}\|\tilde{W}_i\| \le \max\{\chi_{1i}, \chi_{2i}\}$.

For the tracking error the two thresholds are stated on *different functionals* and must not be merged: $\hat{\chi}_{2i}$ bounds the pointwise second moment, whereas $\hat{\chi}_{1i}$ bounds the windowed first moment $\mathbb{E}_0\int_{t-T}^{t}\|\delta_i\|\,d\tau$. They are related in one direction only: by Jensen's and Cauchy–Schwarz inequalities,

$$\Big(\mathbb{E}\int_{t-T}^{t} \|\delta_i(\tau)\|\,d\tau\Big)^2 \le T\int_{t-T}^{t} \mathbb{E}\|\delta_i(\tau)\|^2\,d\tau, \tag{75}$$

so a bound on the windowed first moment does not by itself bound $\mathbb{E}\|\delta_i(t)\|^2$ at the endpoint. The ultimate bound on the tracking error is therefore the pointwise one, $\hat{\chi}_{2i}$, obtained on $\{\xi_i = 1\}$ – the regime in which $\|\delta_i\|$ is growing, and hence the regime that determines the ultimate bound – while $\hat{\chi}_{1i}$ certifies that the windowed error cannot remain large on the contracting regime. Because $\{\xi_i(t) = 1\}$ is precisely the event on which $\|\delta_i\|$ increases over the window, $\mathbb{E}\|\delta_i(t)\|^2$ can exceed $\hat{\chi}_{2i}$ only on $\{\xi_i(t) = 0\}$, where it is non-increasing over the window; hence $\hat{\chi}_{2i}$ is an ultimate bound for the unconditional second moment as well.

Mean-square uniform ultimate boundedness of the augmented state and the weight estimation error is thus established: $\mathbb{E}\|\delta_i\|^2$ is ultimately bounded by $\hat{\chi}_{2i}$ and $\mathbb{E}\|\tilde{W}_i\|$ by $\max\{\chi_{1i}, \chi_{2i}\}$, thresholds that reduce to their deterministic counterparts as $c_{\sigma_i}, c_{V_i} \to 0$, i.e. as $\Sigma_i \to 0$.

Near-optimality. The remaining claims of the theorem follow from the bound on $\tilde{W}_i$ and Assumption 4. From (25) and (30), $V_i^* - \hat{V}_i = \tilde{W}_i^T \phi_i + \epsilon_i$, so

$$\mathbb{E}\|V_i^* - \hat{V}_i\|^2 \leq 2b_{\phi_i}^2\, \mathbb{E}\|\tilde{W}_i\|^2 + 2b_{\epsilon_i}^2 \triangleq \epsilon_{V_i}. \tag{76}$$

Likewise, subtracting (32) from (20) and (33) from (21) and using $\nabla V_i^* = \nabla \phi_i^T W_i + \nabla \epsilon_i$,

$$\mathbb{E}\|u_i^* - \hat{u}_i\|^2 \leq \tfrac{1}{2}(b_i + d_i)^2\, \|R_{ii}^{-1}\|\, \Gamma_{i_{max}} \left(b_{\nabla\phi_i}^2\, \mathbb{E}\|\tilde{W}_i\|^2 + b_{\nabla\epsilon_i}^2\right) \triangleq \epsilon_{u_i}, \tag{77}$$

using $\|R_{ii}^{-1} g_i^T v\|^2 \leq \|R_{ii}^{-1}\|\, v^T \Gamma_i v \leq \|R_{ii}^{-1}\| \Gamma_{i_{max}} \|v\|^2$; analogously $\mathbb{E}\|\vartheta_i^* - \hat{\vartheta}_i\|^2 \leq \epsilon_{\vartheta_i}$ with $R_{ii}^{-1}, \Gamma_{i_{max}}, \frac{1}{2}$ replaced by $T_{ii}^{-1}, \Lambda_{i_{max}}, \frac{1}{2\gamma^2}$. Each of $\epsilon_{V_i}, \epsilon_{u_i}, \epsilon_{\vartheta_i}$ is therefore small whenever $\max\{\chi_{1i}, \chi_{2i}\}$ and the NN approximation errors are small; they shrink to the deterministic residual, not to zero, as $\Sigma_i \to 0$, since the approximation errors $b_{\epsilon_i}, b_{\nabla\epsilon_i}$ persist and vanish only as $L \to \infty$. The proof is thus complete.■

□

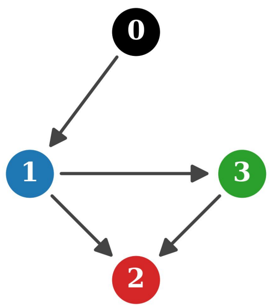
Corollary 1 (Model-free case)**.** *Let the dynamics (11) be* completely unknown*: neither the drift f_{ei}, nor the input channels g_i, k_i, nor the diffusion Σ_i is available. Suppose the behaviour signals $u_j = \hat{u}_j + e_j$ are applied with bounded exploration e_j, the policies and cross-channel maps are parameterised as in (49), and the weights $\mathcal{Z}_i$ of (50) are updated by the adaptive law (35) under the substitution (53). If the enlarged rank condition of Section 4.3 holds, then $\mathbb{E}\|\delta_i(t)\|^2$ and $\mathbb{E}\|\tilde{\mathcal{Z}}_i(t)\|^2$, $\tilde{\mathcal{Z}}_i \triangleq \mathcal{Z}_i^* - \mathcal{Z}_i$, are mean-square UUB with the thresholds $\hat{\chi}_{1i}, \hat{\chi}_{2i}, \chi_{1i}, \chi_{2i}$ of Theorem 1, in which ϵ_{max} is replaced by the corresponding bound on the off-policy residual of (48) and $\lambda_{min}(\Xi_i), \lambda_{max}(\Xi_i)$ by the extreme eigenvalues of the enlarged Gram matrix built from Λ_i.*

Proof. The proof of Theorem 1 uses the adaptive law (35) only through the error dynamics (43), and (43) only through three properties: (i) the Gram matrix Ξ_i is positive semidefinite, with $\lambda_{min}(\Xi_i) > 0$ guaranteed by the rank condition; (ii) the residual term satisfies $\mathbb{E}\|\epsilon_{B_i}\|^2 \leq \epsilon_{max}^2$; and (iii) the stabilizing term is bounded, $\|\Theta_i\| \leq \theta_{max_i}$. None of the three refers to the *contents* of the regressor. Under (53), (i) holds by the enlarged rank condition, (ii) holds because (48) differs from (27) only by terms that are exact under (46)–(47), so its residual is again the NN approximation error of (29) augmented by the actor and cross-channel approximation errors – all bounded in mean square on Ω by Assumption 4 applied to the additional bases $\psi_{u_i}, \psi_{\vartheta_i}, \psi_{ij}$ – and (iii) holds as established after (55). The Lyapunov argument of Theorem 1 therefore applies verbatim with $\tilde{W}_i$ replaced by $\tilde{\mathcal{Z}}_i$.■ □

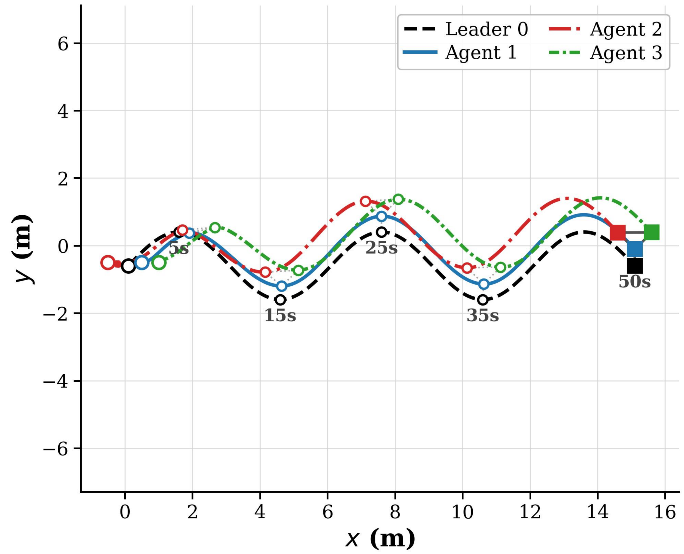
5 Simulation results

In this section, simulations are conducted to validate the performance of the proposed control strategy. The scenario involves a multi-WMR system, where a virtual leader WMR (indexed by 0) guides the formation. The information flow among WMRs is depicted in the directed graph shown in Fig. 1a.

The desired trajectory deviation are set to $\Delta_1 = [0, -0.5, 0]^T, \Delta_2 = [0.5, -1, 0]^T, \Delta_3 = [-0.5, -1, 0]^T$. The initial position of WMR are respectively $q_0 = [0.1, -0.6, 0.808]^T, q_1 =$



(a) Communication graph.



(b) Trajectories of relative motion of the WMRs, with the rigid formation edges redrawn at five time snapshots (5, 15, 25, 35, 50 s) to show the formation actually being carried along the path rather than only its start and end.

Figure 1: Communication topology and the resulting formation trajectory.

$[0.5, -0.5, 0]^T, q_2 = [-0.5, -0.5, 0]^T, q_3 = [1, -0.5, \pi/6]^T$. The initial velocity vectors of the WMRs are set to $\eta_1 = \eta_2 = \eta_3 = [0, 0]^T$, and the simulation horizon is 50 s with step size 0.01 s.

The virtual leader traces a *serpentine* (sinusoidal S-curve) reference path

$$x_0(t) = x_0(0) + v_x t, \qquad y_0(t) = y_0(0) + A\sin(\omega t), \tag{78}$$

which alternates straight and curved segments and reverses its curvature twice per period, so that the reference is genuinely time-varying rather than a steady-state circular motion. The corresponding unicycle velocity reference is obtained from $v_0 = \sqrt{\dot{x}_0^2 + \dot{y}_0^2}$ and $\omega_0 = (\ddot{y}_0\dot{x}_0 - \ddot{x}_0\dot{y}_0)/(\dot{x}_0^2 + \dot{y}_0^2)$, giving

$$\eta_0 = \begin{bmatrix} v_0 \\ \omega_0 \end{bmatrix} = \begin{bmatrix} \sqrt{v_x^2 + \big(A\omega\cos(\omega t)\big)^2} \\ \dfrac{-A\omega^2 v_x \sin(\omega t)}{v_x^2 + \big(A\omega\cos(\omega t)\big)^2} \end{bmatrix}$$

where $v_x = 0.3$ m/s, $A = 1$ m and $\omega = \pi/10$ rad/s, yielding 2.5 complete periods over the simulation horizon, $v_0 \in [0.300, 0.434]$ m/s and $\omega_0 \in [-0.329, 0.329]$ rad/s. The leader heading is initialised on the path tangent, $\theta_0(0) = \arctan(A\omega/v_x) \approx 0.808$ rad, so that integrating $\dot{q}_0 = H_0(q_0)\eta_0$ reproduces (78) exactly. Note that ω_0 here oscillates about zero instead of being constant, which excites the critic regressor more richly than a circular reference and makes the rank requirement of Condition 1 easier to meet: the recorded history stacks reach $\mathrm{rank}(Z_i) = L = 15$ for all three agents.

The NNs activation functions are selected as

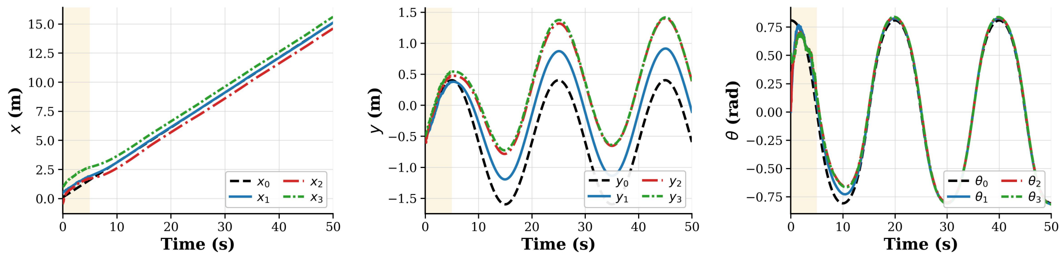
$$\phi_i(\delta_i) = [\delta_{xi}^2, \delta_{xi}\delta_{yi}, \delta_{xi}\delta_{\theta_i}, \delta_{xi}\delta_{vi}, \delta_{xi}\delta_{\omega_i}, \delta_{yi}^2, \delta_{yi}\delta_{\theta_i}, \delta_{yi}\delta_{vi}, \delta_{yi}\delta_{\omega_i}, \delta_{\theta_i}^2, \delta_{\theta_i}\delta_{vi}, \delta_{\theta_i}\delta_{\omega_i}, \delta_{vi}^2, \delta_{vi}\delta_{\omega_i}, \delta_{\omega_i}^2].$$

and the initial NNs weight vectors are set to $\hat{W}_i(0) = 0.5 \cdot \mathbf{1}_{15}$ for all three agents. The learning rates are $\alpha_i = 0.01$ and $\beta_i = 0.01$, $Q_{ii}(e_i) = e_i^T Q_i e_i$, $Q_i = diag[1, 1, 1, 1, 1]$, $T_{ii} = T_{ij} = I$, and $R_{ii} = R_{ij} = 0.3I$ – recall that the control and disturbance laws (32)–(33) act through R_{ii}^{-1}, T_{ii}^{-1}, so this weight corresponds to a feedback gain $R_{ii}^{-1} \approx 3.33I$, not $0.3I$. A sweep of the control-penalty weight over $R_{ii} = R_{ij} = cI$, $c \in \{1, 0.7, 0.5, 0.4, 0.3\}$ – a *smaller* c giving a *larger* gain $R_{ii}^{-1} = c^{-1}I$ – showed the post-transient tracking error and cost shrinking as c is reduced from 1 toward 0.4, after which the trend is no longer monotone and the closed loop becomes numerically unstable below $c \approx 0.28$. We keep $c = 0.3$, close enough to this margin that the two safety mechanisms of (35) are actually exercised: removing either the stabilizing term $\xi_i\Theta_i$ or the weight adaptation destabilizes the closed loop within about three seconds (Sec. 5.3), whereas at looser gains (e.g. $c = 0.4$–0.5) both ablated variants remain stable and that demonstration is not visible. Varying Q_i or T_{ii}, T_{ij} over the same range left the closed loop virtually unchanged, since only R_{ii}, R_{ij} enter the control law u_i^r directly. The selection of physical parameters of each WMR model is chosen as $r_i = 0.05$m, $l_i = 0.25$ m, $m_1 = 10$kg, $m_2 = m_3 = 5$kg. The history stack of Condition 1 retains $l = 20 > L$ samples per agent, admitted at a randomized interval of 0.15–0.35 s (mean 0.25 s) and evicted first-in-first-out once full, so that the stored regressors are drawn from a rolling recent window rather than clustering in one transient; the mean admission rate is chosen so the stack finishes filling within the persistent-excitation window below, rather than on an unrelated, independent timescale. $\mathrm{rank}(Z_i) = L = 15$ is maintained throughout for all three agents.

The evolution of the position states x_i, y_i, and θ_i for robots 1, 2, 3, along with the virtual leader 0, are illustrated in Figs. 2a–2c. The learning progress of the critic neural network weights for each agent is presented in Figs. 3a–3c: all fifteen weights of each agent move together, since they are updated from the same regressor stack, and settle to constant values once that stack fills, by $t \approx 4.7$ s, coinciding with the end of the persistent-excitation window; they remain flat for the rest of the horizon. Furthermore, the 2D trajectories of both the leader and the followers are depicted in Fig. 1b, confirming that the proposed online learning-based control scheme enables the multi-agent system to achieve optimal synchronization.

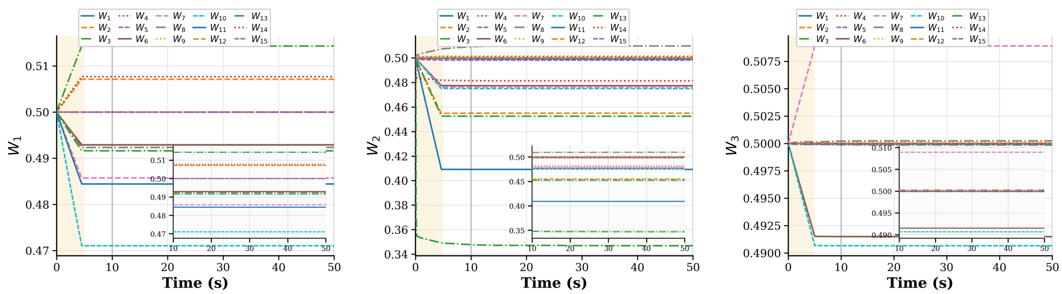
Quantitatively, the formation tracking errors $\|[\delta_{xi}, \delta_{yi}]\|$ decay from 0.57, 1.12 and 0.50 m at $t = 0$, though not all agents settle at the same rate: agents 1 and 3 are down to 0.03 m and 0.01 m by $t = 25$ s, while agent 2 – whose initial error is more than twice the others' and which sits in the middle of the communication graph, coupled to both – traverses a slower, extended transient and does not reach its own residual band until $t \approx 28$–30 s. From $t = 30$ s onward, all three agents remain within a common residual band of roughly 0.00–0.05 m rather than converging to zero. This residual is the expected signature of the mean-square UUB result of Theorem 1: under persistent process noise the errors cannot vanish, and the ultimate bound $\hat{\chi}_{2i}$ of (72) is non-zero for $\Sigma_i \neq 0$. Consistent with this bound trading off against the feedback gain through $\Gamma_{i,max}$, the sweep above shows the residual growing monotonically as c increases toward 1 (i.e. as the gain R_{ii}^{-1} weakens), and at the operating point $c = 0.3$ the residual shows only a mild, low-amplitude wobble with no growth over the horizon. These results demonstrate that the proposed approach drives the system states to track a genuinely time-varying reference under process noise, thereby validating the efficacy of the developed method.

The individual neighborhood tracking errors e_x, e_y, and e_θ that feed the critic NN are shown in Figs. 4a–4c, confirming the same transient decay and bounded residual behavior discussed above at the component level; each signed component still fluctuates about zero at this small amplitude rather than converging to it exactly, consistent with a non-zero (if now much smaller) mean-square ultimate bound. The corresponding position-error norm $\|\delta_i\| = \sqrt{e_x^2 + e_y^2}$, shown in Fig. 4d, makes this UUB behavior easier to read at a glance: it is non-negative by construction, decays over the transient (monotonically for agents 1 and 3, with agent 2's slower shoulder visible until $t \approx 30$ s as discussed above), and then settles into the 0.00–0.05 m residual band quoted above. The estimated value functions $\hat{V}_i = \hat{W}_i^T \phi_i(\delta_i)$, shown in Fig. 5a, start from the transient peak induced by the initial tracking error and decay to a small residual as the critic weights converge, consistent with the near-optimality bound $\mathbb{E}\|V_i^* - \hat{V}_i\|^2 < \epsilon_{V_i}$ of Theorem 1. The resulting control inputs v_i and ω_i are shown in Figs. 5b–5c: the large-amplitude fluctuations over $[0, 5]$ s are the injected persistent-excitation exploration noise used to populate the history stack (Condition 1), after which the control smoothly tracks the leader's serpentine reference velocity without chattering.



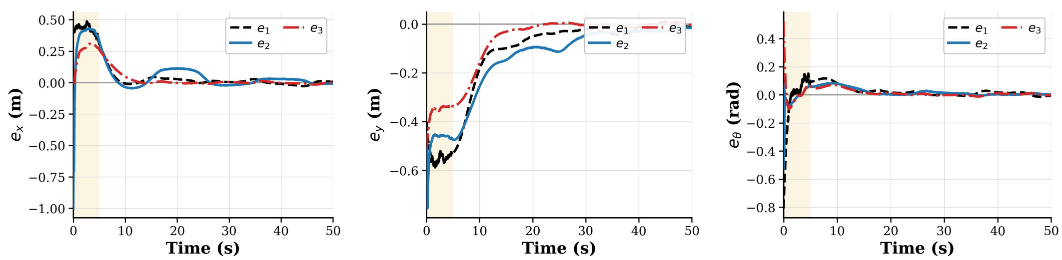
(a) x_i $(i = 1, 2, 3), x_0$ (b) y_i $(i = 1, 2, 3), y_0$ (c) θ_i $(i = 1, 2, 3), \theta_0$

Figure 2: Position and heading trajectories of the leader and the three followers.

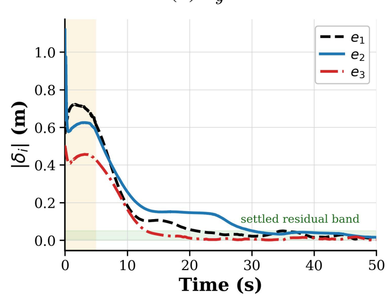


(a) Agent 1 (b) Agent 2 (c) Agent 3

Figure 3: Weight update trajectories of the critic NNs, agents 1–3.



(a) e_x (b) e_y (c) e_θ

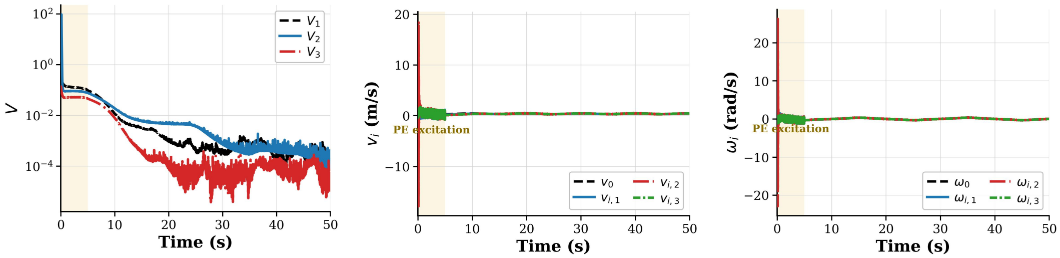


(d) Position-error norm $\|\delta_i\|$

Figure 4: Neighborhood tracking errors e_x, e_y, e_θ $(i = 1, 2, 3)$, and the resulting position-error norm.

5.1 Sensitivity to the process-noise intensity

Theorem 1 claims that the mean-square ultimate bound $\hat{\chi}_{2i}$ reduces exactly to the deterministic (pathwise) UUB bound as $\Sigma_i \to 0$, and grows with the noise intensity otherwise. To check this directly rather than only asserting it, we repeat the full experiment above – same topology, reference, initial conditions, learning rates, and $R_{ii} = 0.3I$ – at $\sigma_\eta \in \{0, 0.03, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6\}$



(a) Value functions $\hat{V}_i$ (b) Control input v_i (c) Control input ω_i

Figure 5: Estimated value functions and the resulting control inputs v_i, ω_i $(i = 1, 2, 3)$.

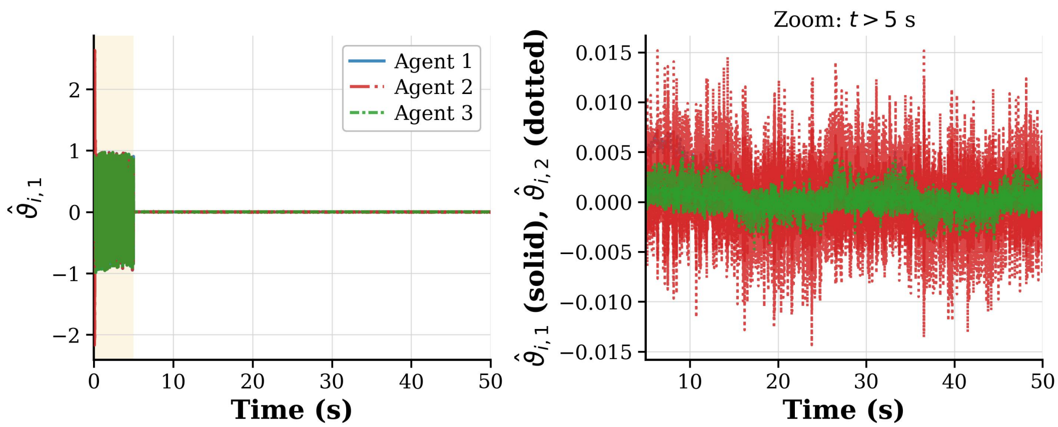


Figure 6: Estimated worst-case disturbance policy $\hat{\vartheta}_i$ (33), dynamic-channel component 1. Left: full horizon, dominated over $[0, 5]$ s by the same injected persistent-excitation noise seen in Figs. 5b–5c – shared identically across the three agents, which is why agents 1 and 3 (of comparable magnitude) are visually indistinguishable here. Right: zoom on $t > 5$ s, both dynamic-channel components, showing the converged worst-case policy settling to a small, noise-driven fluctuation about zero rather than a fixed value, consistent with $\vartheta_i \to 0$ as $\delta_i \to 0$.

(recall $\sigma_\eta = 0.03$ is the value used in every other figure in this section), keeping the seed and hence the realized Brownian path fixed up to the rescaling by σ_η. Figure 7 reports the settled residual band, mean and max of $\|\delta_i\|$ over $t > 30$ s, against σ_η, together with the full worst-agent trajectories.

Three observations follow. First, $\sigma_\eta = 0$ does *not* give zero residual: the settled band is mean $=$ 0.022 m, consistent with the theorem's own remark that the near-optimality constants $\epsilon_{V_i}, \epsilon_{u_i}, \epsilon_{\vartheta_i}$ persist in the deterministic limit and vanish only as the critic basis is refined ($L \to \infty$), not as $\Sigma_i \to 0$ alone. Second, the residual grows with σ_η essentially monotonically – from mean $= 0.022$ m at $\sigma_\eta = 0$ to 0.057 m at $\sigma_\eta = 0.6$, roughly a $2.6\times$ increase, and more sharply so in the max statistic (0.049 m $\to 0.241$ m) – exactly the qualitative trend $\hat{\chi}_{2i}$ predicts through its c_{V_i} term, which is itself increasing in Σ_i. Third, and visible in Fig. 7(right), the *transient* (dominated by the initial tracking error, not the noise) is essentially unaffected by σ_η: all eight traces overlap almost exactly for $t < 25$ s and only separate in the settled regime, confirming that σ_η acts on the ultimate bound and not on the transient decay rate, as the theory requires. Beyond $\sigma_\eta \approx 0.65$ the closed loop becomes numerically unstable, consistent with Assumptions 1 and 4 requiring the diffusion to remain within the regime over which the SDE is well-posed and the NN bases and their derivatives stay bounded.

5.2 Robustness to an exogenous disturbance

The results above already exercise the disturbance channel, since the worst-case policy $\hat{\vartheta}_i$ of (33) is applied to the plant throughout as the self-play adversary (Fig. 6). This is not, however, a test against a disturbance the controller did not itself generate. To check the disturbance-attenuation condition

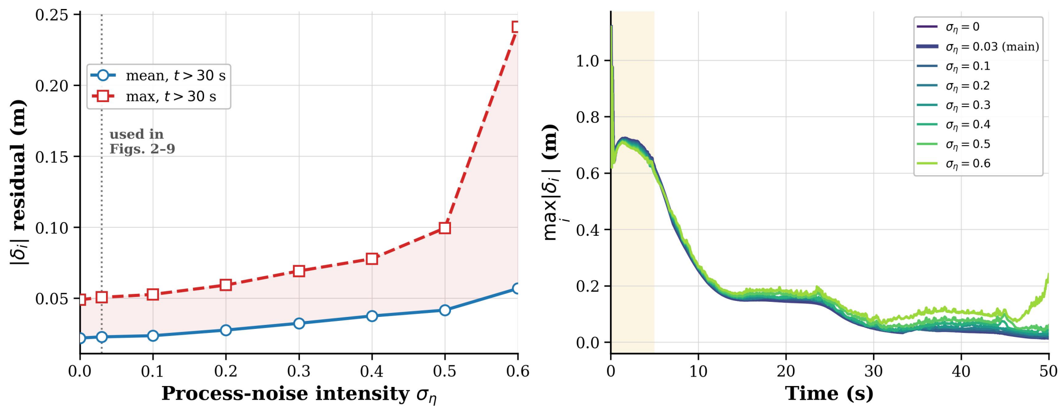


Figure 7: Sensitivity of the settled residual to the process-noise intensity σ_η. Left: mean and max of $\|\delta_i\|$ over $t > 30$ s vs. σ_η; the dotted line marks $\sigma_\eta = 0.03$, the value used in every other figure in this section. Right: the full worst-agent trajectory $\max_i \|\delta_i(t)\|$ for each σ_η, showing the transient is essentially noise-independent while the settled band widens with σ_η.

(14) against a genuinely exogenous signal, we inject a bounded torque disturbance $d_{\text{ext}}(t)$ – a low-frequency oscillation (0.3 rad/s, amplitude 1.5 N·m and half that on the two wheel channels) plus a 1 N·m step at $t = 20$ s – into the dynamic channel of every agent, identically, on top of (not instead of) the existing self-play disturbance, and repeat the experiment with every other setting unchanged. Figure 8 compares the resulting worst-agent tracking error against the undisturbed baseline and shows the applied disturbance itself.

The tracking error with and without d_{ext} is graphically indistinguishable for $t < 35$ s and separates only modestly thereafter: the settled residual (mean over $t > 25$ s) rises from 0.026 m to 0.045 m, and the disturbed trajectory does not grow further after the step at $t = 20$ s or the oscillation's subsequent cycles – it remains bounded, as (14) requires. Quantitatively, evaluating the two sides of (14) directly from the disturbed run over $t > 25$ s (past the initial transient, so the $\beta_i(\delta_i(0))$ term is not dominant) gives $\sum_i \int \|z_i\|^2 \, dt = 1.13$ against $\sum_i \int \|\omega_i\|^2 \, dt = 450.3$, a ratio of 0.0025 – comfortably below $\gamma^2 = 1$. The margin is large rather than tight, which is expected: as Remark 2 notes, γ is deliberately chosen conservatively in the absence of a closed-form optimal value, and the design weights $R_{ii} = 0.3I$ were chosen (Sec. 5, first paragraph) for the tracking/stability trade-off of Sec. 5.3, not to tighten this particular margin.

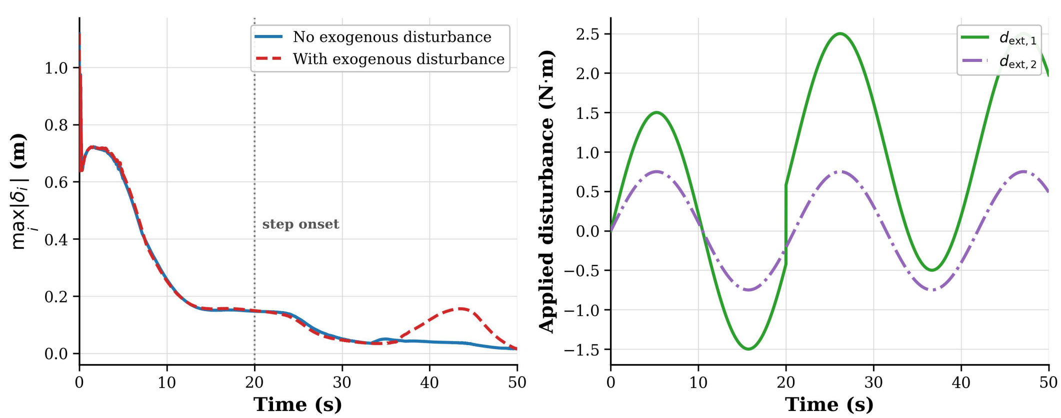


Figure 8: Robustness to a genuinely exogenous disturbance. Left: worst-agent position-error norm with and without the applied disturbance d_{ext} (vertical line: step onset). Right: the applied disturbance itself, on the two components of the dynamic channel.

5.3 Contribution of the learning component

A natural question is how much of the reported performance is actually attributable to the reinforcement-learning machinery, as opposed to the model-based terms v_a, τ_a or to the particular initial weight guess. We therefore repeat the experiment with individual components of the scheme disabled, keeping every other setting (topology, reference, noise realisation, seed) fixed. The results are summarised in Table 1 and Fig. 9.

Table 1: Ablation of the proposed scheme. $J = \sum_i \int_5^{50} (e_i^T Q_i e_i + u_i^{rT} R_{ii} u_i^r)\, dt$ is evaluated after the excitation window; $\|\bar{\delta}\|$ is the mean position-error norm over $t > 25$ s.

Variant	J	$\|\bar{\delta}\|$ (m)
Proposed (full)	4.38	0.026
Without Bellman-residual / replay terms	4.49	0.025
Without stabilizing term $\xi_i \Theta_i$	diverges	—
Critic weights frozen at $\hat{W}_i(0)$	diverges	—
Without learned policy u_i^r	1.5×10^8	14.11

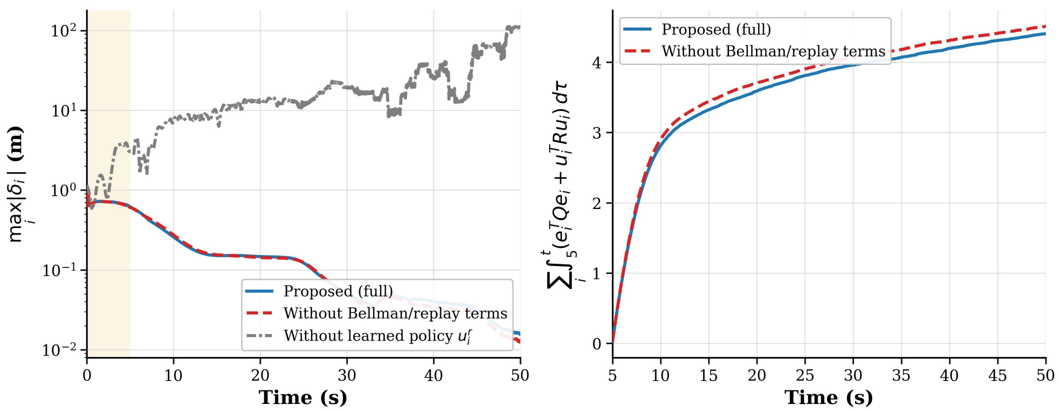


Figure 9: Contribution of the learning component. Left: worst-agent formation-error norm; removing the learned policy u_i^r leaves the model-based terms unable to stabilise the formation, whereas the full scheme and the variant without the Bellman/replay terms are graphically indistinguishable. Right: accumulated cost after the excitation window, where the two curves separate slightly — the Bellman-residual and experience-replay terms lower the cost functional by a modest $\sim 2\%$ at equal tracking accuracy.

Three points follow. First, the learned policy is not a refinement of an already-working controller: with u_i^r removed the neighbourhood error grows past 14 m, so the model-based feedforward alone does not stabilise the formation. Second, the two halves of the tuning law (35) play distinct roles. Removing the stabilizing term $\xi_i \Theta_i$, or freezing the weights at $\hat{W}_i(0) = 0.5 \cdot \mathbf{1}$, makes the closed loop diverge within about three seconds — unsurprisingly, since that initial guess assigns equal weight to every quadratic monomial and hence induces an indefinite $\hat{V}_i$ and a non-stabilising policy at the feedback gain R_{ii}^{-1} used here. The stabilizing term is what carries the state out of that region; as noted above, this demonstration relies on operating close to the stability margin identified by the R_{ii} sweep; at a looser gain neither ablated variant diverges. Third, and most relevant to the claim of *optimality*, the Bellman-residual and experience-replay terms leave the tracking error essentially unchanged (0.026 versus 0.025 m) while reducing the cost functional from 4.49 to 4.38, a modest but consistent 2.4% improvement, split roughly evenly between the state term (3.32 versus 3.23) and the control-effort term (1.17 versus 1.16) rather than concentrated in either one. The effect is small because, at this operating point, even the frozen initial weights already yield a policy close enough to stabilizing that little further improvement is available before the residual band imposed by the process noise (Theorem 1) is reached; nonetheless the direction of the effect is exactly what the theory predicts: $\xi_i\Theta_i$ secures the mean-square UUB guarantee of Theorem 1 and is responsible for the difference between stability and divergence, while the reinforcement-learning terms drive $\hat{W}_i$ towards the solution of the coupled HJI equations and thus towards the optimal policy, lowering the cost at equal tracking accuracy. It also illustrates why tracking-error plots alone are a poor diagnostic for this class of scheme: the error is not the quantity being minimised, and a scheme can track well with a policy that is still measurably sub-optimal in cost.

6 Conclusions

This study introduces a novel control strategy for achieving synchronization in multi-wheel mobile robot systems subject to both stochastic process noise and worst-case external disturbances. The proposed framework incorporates elements of distributed control, optimal control, and stochastic differential game theory to tackle the multi-agent zero-sum graphical game problem under Itô stochastic dynamics. The control scheme is developed based on a set of coupled stochastic HJI equations – which gain a genuine Itô-correction term over their deterministic counterpart and reduce to them exactly as the diffusion intensity vanishes – and the resulting solution ensures that the global synchronization error remains bounded in the L_2-norm in the mean-square sense.

Furthermore, an innovative online reinforcement learning (RL) algorithm is developed based on neural networks (NNs) to achieve optimal formation tracking under process noise. Each agent employs a single-layer NN to approximate its value function, while a concurrent learning mechanism utilizing historical data is introduced to relax the need for traditional persistent excitation (PE) conditions. This facilitates the adaptive tuning of NN parameters in real time, and the resulting closed-loop tracking error and weight estimation error are shown to be ultimately uniformly bounded in the mean-square sense, with an explicit noise-dependent ultimate bound.

Finally, Section 4.3 removes the remaining model dependence. Whereas the IRL-based Bellman formulation already eliminates the drift, the policies and the stabilizing term still evaluate the input channels g_i, k_i explicitly. An off-policy reformulation, in which the plant is driven by exploration-perturbed behaviour signals while the target policies are evaluated and improved, lumps the unknown products $\nabla V_i^T g_i$ and $\nabla V_i^T k_i$ into the agent's own policies through the stationarity conditions, and identifies the graphical-game cross-channel terms $\nabla V_i^T g_j$ from data. Because the process noise of (5) enters additively rather than through the control channel, no diffusion-dependent parameter has to be identified – a simplification unavailable in the linear stochastic case [21], where the control-dependent diffusion forces an extra lumped unknown. The extension is a change of regressor rather than a change of algorithm, so the mean-square UUB guarantee carries over unchanged (Corollary 1), at the price of a correspondingly richer excitation requirement.

Future work will address the sample complexity of the enlarged excitation condition, which grows with the neighbourhood size $|\mathcal{N}_i|$, and the robustness of the scheme to the estimation errors incurred when the conditional expectations in (48) are approximated from finitely many sample paths, along the lines of the input-to-state stability analysis of [21].

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Appendix

A Appendix

Proof of Lemma 1. Let $\bar{e}_i \triangleq [(p_i + \Delta_i - p_0)^T, (\eta_i - \eta_0)^T]^T \in \mathbb{R}^5$ denote the actual, leader-relative formation tracking error of agent i under the original dynamics (5): driving $\mathbb{E}\|\bar{e}_i(t)\|^2 \to 0$ (or keeping it mean-square bounded) is precisely the objective of the distributed H_∞ optimal *formation tracking* problem stated for (5). Write $\bar{e} \triangleq [\bar{e}_1^T, \ldots, \bar{e}_n^T]^T$.

Expanding the definitions of $\delta_{pi}, \delta_{\eta i}$ preceding (7),

$$\delta_{pi} = \sum_{j \in \mathcal{N}_i} a_{ij} \big[(p_i + \Delta_i - p_0) - (p_j + \Delta_j - p_0)\big] + b_i(p_i + \Delta_i - p_0) = (d_i + b_i)\, \bar{e}_{pi} - \sum_{j \in \mathcal{N}_i} a_{ij}\, \bar{e}_{pj},$$

which is exactly row i of $(\mathcal{D} + \mathcal{B} - \mathcal{A})\bar{e}_p = (\mathcal{L} + \mathcal{B})\bar{e}_p$, where $\bar{e}_{pi} \triangleq p_i + \Delta_i - p_0$; the identical computation for $\delta_{\eta i}$ with $\bar{e}_{\eta i} \triangleq \eta_i - \eta_0$ gives row i of $(\mathcal{L} + \mathcal{B})\bar{e}_\eta$. Stacking both channels,

$$\delta = T\bar{e}, \qquad T \triangleq (\mathcal{L} + \mathcal{B}) \otimes I_5, \tag{79}$$

with δ as defined before (10).

T is invertible. By the standing assumptions of Section 2 the communication graph is strongly connected and at least one follower has $a_{i0} = b_i > 0$; under these conditions $\mathcal{L} + \mathcal{B}$ is a nonsingular M-matrix, a standard fact in the leader-follower consensus literature underlying graphical-game formulations such as [24] (every eigenvalue of $\mathcal{L} + \mathcal{B}$ has strictly positive real part, since $\mathcal{L}$ has a simple zero eigenvalue with eigenvector $\mathbf{1}$ that is destroyed once $\mathcal{B} \neq 0$ couples at least one node to the leader through a strongly connected graph). Hence $\mathcal{L} + \mathcal{B}$, and therefore $T = (\mathcal{L} + \mathcal{B}) \otimes I_5$, is invertible, with

$$\underline{\kappa}\|\bar{e}\| \;\le\; \|\delta\| \;\le\; \bar{\kappa}\|\bar{e}\| \tag{80}$$

for constants $0 < \underline{\kappa} \le \bar{\kappa} < \infty$ given by the extreme singular values of T and T^{-1}.

T is exact, not merely infinitesimal. Because T is a *constant* matrix, $\delta(t) = T\bar{e}(t)$ holds pathwise at every t, not only to first order: differentiating (79) requires no Itô correction (unlike a nonlinear change of variables), so $\bar{e}$ driven by (5) and δ driven by (10) are related by the *same* linear map T at every sample path and every time, for matching initial conditions $\delta(0) = T\bar{e}(0)$ and identical control, disturbance and Brownian-motion realizations (u, ϑ, W). In particular (80) holds at every t along every trajectory, so $\mathbb{E}\|\delta(t)\|^2 \to 0$ (or remains ultimately bounded by χ) if and only if $\mathbb{E}\|\bar{e}(t)\|^2 \to 0$ (respectively, remains ultimately bounded by a constant in $[\underline{\kappa}^{-2}\chi, \bar{\kappa}^{-2}\chi]$); i.e. admissibility of a control law $u_i \in \Psi(\Omega)$ for (10) in the sense of Definition 1 is equivalent to mean-square stabilization of the original formation error $\bar{e}_i$ under (5).

The cost and the saddle point transfer unchanged. The performance index (13) is, by definition, a functional of $\delta_i(\cdot)$ and of the *same* input signals $u_i, u_{-i}, \vartheta_i, \vartheta_{-i}$ that drive both (5) and (10); T acts only on the state and does not touch the input channels g_i, k_i, which enter (10) exactly as they enter (5) (11) after the same computation used above. Consequently, for any admissible inputs, $J_i(\delta_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$ evaluated along a trajectory of (10) equals $J_i(T\bar{e}_i(0), u_i, u_{-i}, \vartheta_i, \vartheta_{-i})$ evaluated along the *corresponding* trajectory of (5) (same inputs, same noise, initial conditions related by $\delta_i(0) = (T\bar{e}(0))_i$), because the two trajectories coincide under T by the previous paragraph. The zero-sum game $\min_{u_i} \max_{\vartheta_i} J_i$ therefore has the same value and the same saddle-point input trajectories (u_i^*, ϑ_i^*) whether posed on (10) or, after composing with the fixed invertible map T, on (5).

Combining the three paragraphs above: a policy solves the distributed H_∞ optimal control problem of Section 3 for (10) if and only if it solves, in the mean-square sense, the distributed H_∞ optimal formation tracking problem for (5), which is the claim of Lemma 1. $\square$